ALGEBRA II · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

84 question sets • 2525 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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Write complex numbers in the form a + bi

MAIN	Algebra II main question	**HSN-CN.A.1**
BACKWARD 1	Simplify square roots of positive numbers	**HSN-RN.A.2**
BACKWARD 2	No real number squares to a negative number	**8.EE.A.2**
BACKWARD 3	Divide each term of a sum by a number	**6.EE.A.3**
FORWARD 1	Use equality of complex numbers to find unknown real values	**HSN-CN.A.1**
FORWARD 2	Represent complex numbers as points in the complex plane	**HSN-CN.B.4**

SET 1 • HSN-CN.A.1

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-49)}$ in the form bi, where b is a real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

7i

GRADING NOTE Form required: bi. $\sqrt{(-49)} = \sqrt{49} \cdot \sqrt{(-1)} = 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

Write √(−18) in the form bi with b in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

3i√2 (that is, 3√2 i)

GRADING NOTE Form required: simplest radical form. √(−18) = √9 · √2 · i = 3√2 i. Not accepted: √18 i (not simplified).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

Write $6 - \sqrt{(-20)}$ in the form $a + bi$. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

$6 - 2\sqrt{5}\,i$; real part 6, imaginary part $-2\sqrt{5}$

GRADING NOTE Form required: $a + bi$ with both parts named. $\sqrt{(-20)} = 2\sqrt{5}\,i$. The imaginary part is the real number $-2\sqrt{5}$, not $-2\sqrt{5}\,i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

Write (−4 + √(−75)) / 2 in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

Short answer

CORRECT ANSWER

−2 + (5√3/2)i; real part −2, imaginary part 5√3/2

GRADING NOTE Form required: a + bi with both parts named. √(−75) = 5√3 i; divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

Write (10 + √(−64)) / 4 in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

5/2 + 2i; real part 5/2, imaginary part 2

GRADING NOTE Form required: a + bi with both parts named (5/2 or 2.5 accepted). √(−64) = 8i; 10/4 = 5/2 and 8/4 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 1

Simplify square roots of positive numbers • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify √50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5√2

GRADING NOTE Form required: simplest radical form. 50 = 25 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q1**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

Simplify √72.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6√2

GRADING NOTE Form required: simplest radical form. 72 = 36 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q2**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

Which is √12 in simplest radical form?

A. 4√3

B. 3√2

C. 6√2

D. 2√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2√3

GRADING NOTE $12 = 4 \cdot 3$, so $\sqrt{12} = 2\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q3**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 7$. A square root of a sum is not the sum of the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

Simplify √28.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2√7

GRADING NOTE Form required: simplest radical form. 28 = 4 · 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q5**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 2

No real number squares to a negative number • Grade 8 • Expressions & Equations • 8.EE.A.2

Find all real numbers x with $x^2 = 81$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 9 or x = −9

GRADING NOTE Any order. Both 9^2 and $(-9)^2$ equal 81.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

True or false?

$(-3)^2 = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-3)^2 = (-3)(-3) = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

Is there a real number x with $x^2 = -4$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

Short answer

CORRECT ANSWER

No: the square of any real number is 0 or positive.

GRADING NOTE Any equivalent reason (a positive times a positive and a negative times a negative are both positive).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

Which number has a square equal to 36?

A. 18

B. $\sqrt{-36}$

C. -6

D. -18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −6

GRADING NOTE $(-6)^2 = 36$. $\sqrt{-36}$ is not a real number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(-7)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

Short answer

CORRECT ANSWER

49

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 3

Divide each term of a sum by a number • Grade 6 • Expressions & Equations • 6.EE.A.3

Write (8 + 6√2) / 2 as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

27

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 3√2

GRADING NOTE Any equivalent form. Divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (−6 + 4√5) / 2 as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

28

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−3 + 2√5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

True or false?

$(10 + \sqrt{7}) / 5 = 2 + \sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

29

QUESTION ID

S1-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both terms must be divided: $(10 + \sqrt{7})/5 = 2 + \sqrt{7}/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

(12 − 9√3) / 3 equals

A. 4 − 3√3

B. 4 − 9√3

C. 4 − 3√1

D. 12 − 3√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

30

QUESTION ID

S1-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4 − 3√3

GRADING NOTE 12/3 = 4 and 9√3/3 = 3√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

Write (−4 + 2√3) / (−2) as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

31

QUESTION ID

S1-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2 − √3

GRADING NOTE Any equivalent form. −4/(−2) = 2; 2√3/(−2) = −√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 1

Use equality of complex numbers to find unknown real values • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

x and y are real numbers and (x − 3) + 2yi = 5 − 8i.
Find x and y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

33

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 8, y = −4

GRADING NOTE Form required: both values. Equal complex numbers have equal real parts (x − 3 = 5) and equal imaginary parts (2y = −8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

a and b are real numbers and 3a + (b + 1)i = −12 + 4i. Find a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

34

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = −4, b = 3

GRADING NOTE Form required: both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

True or false?

If x and y are real and x + yi = 7, then y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

35

QUESTION ID

S1-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x + yi = 7 + 0i, so x = 7 and y = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

p and q are real and (p + q) + (p − q)i = 10 + 2i. What is p?

A. 6

B. 5

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

36

QUESTION ID

S1-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE p + q = 10 and p − q = 2 give p = 6 and q = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

m is a real number. For what value of m is $(m^2 - 4) + (m + 3)i$ a real number? What real number is it then?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

37

QUESTION ID

S1-F1-Q5

TYPE

Short answer

CORRECT ANSWER

m = −3; the number is 5

GRADING NOTE Form required: both. The imaginary part m + 3 must be 0, so m = −3 and $m^2 - 4 = 9 - 4 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 2

Represent complex numbers as points in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

In the complex plane the horizontal axis is real and the vertical axis is imaginary. Give the point for −3 + 5i.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

39

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, 5)

GRADING NOTE Form required: an ordered pair (real part, imaginary part).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

Which complex number is represented by the point (4, −2) in the complex plane?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

40

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 − 2i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

Find the distance from 0 to the point 6 + 8i in the complex plane.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

41

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $\sqrt{(6^2 + 8^2)} = 10$ (the modulus $|6 + 8i|$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number lies on the horizontal axis of the complex plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

42

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real number a is a + 0i, the point (a, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

Which number lies on the vertical axis of the complex plane?

A. -7

B. $7 + i$

C. $-7i$

D. $1 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

43

QUESTION ID
S1-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. $-7i$

GRADING NOTE $-7i = 0 - 7i$, the point $(0, -7)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

Add and subtract complex numbers

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Add and subtract integers and decimals	**7.NS.A.1**
BACKWARD 3	Write square roots of negative numbers with i	**HSN-CN.A.1**
FORWARD 1	Find real parameters that make a sum real or imaginary	**HSN-CN.A.2**
FORWARD 2	Add complex numbers as vectors in the complex plane	**HSN-CN.B.5**

SET 2 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Simplify (3 + 5i) + (−7 + 2i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

46

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

−4 + 7i

GRADING NOTE Form required: a + bi. 3 − 7 = −4 and 5i + 2i = 7i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

Simplify (6 − 4i) − (−2 + 9i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

47

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

8 − 13i

GRADING NOTE Form required: a + bi. Subtract both parts: 6 − (−2) = 8 and −4i − 9i = −13i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

Simplify (−1 + √(−16)) + (5 − √(−9)). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

48

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

4 + i

GRADING NOTE Form required: a + bi. (−1 + 4i) + (5 − 3i) = 4 + i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

$z_1 = 2.5 - i$, $z_2 = -1.5 + 3i$ and $z_3 = 4i$. Find $z_1 - z_2 + z_3$ in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

49

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

4 (that is, 4 + 0i)

GRADING NOTE Form required: a + bi or the real number 4. Real: 2.5 + 1.5 = 4; imaginary: −1 − 3 + 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

Find the complex number w for which
(4 − 7i) + w = −2 + 3i.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

50

QUESTION ID
S2-M-Q5

TYPE
Short answer

CORRECT ANSWER
−6 + 10i

GRADING NOTE Form required: a + bi. w = (−2 + 3i) − (4 − 7i) = −6 + 10i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (3 + 5x) + (−7 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

52

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−4 + 7x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

Simplify (6 − 4x) − (−2 + 9x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

53

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8 − 13x

GRADING NOTE Any equivalent form. The minus sign applies to both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

(2 − 3y) − (5 − y) equals

A. −3 − 2y

B. 7 − 4y

C. −3 + 2y

D. −3 − 4y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

54

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −3 − 2y

GRADING NOTE 2 − 5 = −3 and −3y + y = −2y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$(4 + 2t) + (1 - 6t) = 5 + 4t.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

55

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2t - 6t = -4t$, so the sum is $5 - 4t$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (0.5 − x) − (1.5 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

56

QUESTION ID

S2-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−1 − 3x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 2

Add and subtract integers and decimals • Grade 7 • The Number System • 7.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $6 - (-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

58

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-4 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

59

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$-1 - 3 + 4$ equals

A. 8

B. 0

C. -8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

60

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0

GRADING NOTE $-1 - 3 = -4$ and $-4 + 4 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 - (-1.5) = 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

61

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2.5 + 1.5 = 4.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

62

QUESTION ID

S2-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 3

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−16) in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

64

QUESTION ID

S2-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $\sqrt{(-9)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

65

QUESTION ID

S2-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

√(−25) equals

A. −5

B. 25i

C. −5i

D. 5i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

66

QUESTION ID

S2-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5i

GRADING NOTE √(−25) = √25 · i = 5i (the principal root).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{(-1)} = i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

67

QUESTION ID

S2-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of i as the principal square root of −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $-\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

68

QUESTION ID

S2-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−6i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 1

Find real parameters that make a sum real or imaginary • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

k is a real number. For what value of k is (3 + ki) + (k − 5i) a real number? Give that real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

70

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

k = 5; the sum is 8

GRADING NOTE Form required: k and the number. The imaginary part k − 5 must be 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

k is real. For what value of k is
(2k − 6) + 4i − (k + 1 − i) a pure imaginary number
(real part 0)?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

71

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. Real part: 2k − 6 − k − 1 = k − 7 = 0. The sum is then 5i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

True or false?

The sum of a complex number and its conjugate is always 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

72

QUESTION ID

S2-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (a + bi) + (a − bi) = 2a, which is 0 only when a = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

a is real and (a + 3i) − (2 + ai) is real. What is a?

A. 0

B. 2

C. 3

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

73

QUESTION ID

S2-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE Imaginary part 3 − a = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

Explain why the sum of a + bi and a − bi is always a real number when a and b are real.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

74

QUESTION ID

S2-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The imaginary parts b and −b add to 0, leaving 2a, a real number.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q5**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 2

Add complex numbers as vectors in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5

Plot 2 + i and 1 + 3i as points in the complex plane (real axis horizontal). What point represents their sum?

Sum point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

76

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 4), the number 3 + 4i

GRADING NOTE Form required: the point or the number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

The points 0, 2 + i, 1 + 3i and their sum are vertices of a quadrilateral. What kind of quadrilateral is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

77

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A parallelogram

GRADING NOTE Any equivalent answer. Adding complex numbers adds their vectors (parallelogram rule).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

True or false?

Subtracting w from z gives the vector from the point w to the point z.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

78

QUESTION ID
S2-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE z − w is the displacement from w to z.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

What is the distance between the points 5 + 2i and 1 − i in the complex plane?

A. 25

B. √13

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

79

QUESTION ID

S2-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 5

GRADING NOTE |(5 + 2i) − (1 − i)| = |4 + 3i| = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

Find the midpoint of the segment joining $6 - 2i$ and $-2 + 4i$, as a complex number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

80

QUESTION ID

S2-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2 + i$

GRADING NOTE Form required: a + bi. The average $((6 - 2i) + (-2 + 4i))/2 = 2 + i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

Multiply complex numbers and simplify powers of i

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Multiply binomials	**HSA-APR.A.1**
BACKWARD 2	Use $i^2 = -1$ and the cycle of powers of i	**HSN-CN.A.1**
BACKWARD 3	Divide whole numbers to find a remainder	**4.NBT.B.6**
FORWARD 1	Find a real parameter that makes a product real	**HSN-CN.A.2**
FORWARD 2	Divide complex numbers using conjugates	**HSN-CN.A.3**

SET 3 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply (3 + 2i)(4 − 5i). Write the product in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

83

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

22 − 7i

GRADING NOTE Form required: a + bi. $12 - 15i + 8i - 10i^2 = 12 - 7i + 10 = 22 - 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

Simplify $(-2 + i)^2$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

84

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

3 − 4i

GRADING NOTE Form required: a + bi. $4 - 4i + i^2 = 4 - 4i - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

Multiply $(5 - 3i)(5 + 3i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

85

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

34 (that is, 34 + 0i)

GRADING NOTE Form required: $a + bi$ or the real number 34. $25 + 15i - 15i - 9i^2 = 25 + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

Simplify $(1 + i)^3$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

86

QUESTION ID

S3-M-Q4

TYPE

Short answer

CORRECT ANSWER

−2 + 2i

GRADING NOTE Form required: a + bi. $(1 + i)^2 = 2i$, and $2i(1 + i) = 2i + 2i^2 = -2 + 2i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

Simplify $i^{23} \cdot (4 - i)$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

87

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

−1 − 4i

GRADING NOTE Form required: a + bi. 23 = 4 · 5 + 3, so $i^{23} = i^3 = -i$; $-i(4 - i) = -4i + i^2 = -1 - 4i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 1

Multiply binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply (3 + 2x)(4 − 5x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

89

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$12 - 7x - 10x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(-2 + x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

90

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$4 - 4x + x^2$

GRADING NOTE Any equivalent form. Not $x^2 + 4$ (missing the middle term).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

(5 − 3x)(5 + 3x) equals

A. $25 + 9x^2$

B. $10 − 9x^2$

C. $25 − 9x^2$

D. $25 − 30x − 9x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

91

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $25 − 9x^2$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$(1 + x)^2 = 1 + x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

92

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(1 + x)^2 = 1 + 2x + x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

Multiply $x(4 - x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

93

QUESTION ID

S3-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$4x - x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 2

Use $i^2 = -1$ and the cycle of powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

95

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. Definition of i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

Simplify i^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

96

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-i$

GRADING NOTE Any equivalent form. $i^3 = i^2 \cdot i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

Simplify i^4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

97

QUESTION ID

S3-B2-Q3

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. $i^4 = (i^2)^2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-10i^2 = -10.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

98

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^2 = -1$, so $-10i^2 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

Which equals 1?

A. i^8

B. i^9

C. i^3

D. i^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

99

QUESTION ID

S3-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. i^8

GRADING NOTE Powers of i repeat every 4; 8 is a multiple of 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 3

Divide whole numbers to find a remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find the remainder when 23 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

101

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 23 = 4 · 5 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

Find the remainder when 50 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

102

QUESTION ID

S3-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 50 = 4 · 12 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

What is the remainder when 36 is divided by 4?

A. 0

B. 4

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

103

QUESTION ID

S3-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0

GRADING NOTE $36 = 4 \cdot 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

True or false?

The remainder when 101 is divided by 4 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

104

QUESTION ID

S3-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $101 = 4 \cdot 25 + 1$, so the remainder is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

Write 39 as 4q + r with 0 ≤ r < 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

105

QUESTION ID

S3-B3-Q5

TYPE

Short answer

CORRECT ANSWER

39 = 4 · 9 + 3

GRADING NOTE Form required: q = 9 and r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 1

Find a real parameter that makes a product real • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

a is a real number. For what value of a is (a + 2i)(3 − i) a real number? Give the product.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

107

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

a = 6; the product is 20

GRADING NOTE Form required: both. (a + 2i)(3 − i) = (3a + 2) + (6 − a)i; the imaginary part 6 − a = 0. Product 18 + 2 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

b is real. For what value of b is (2 + bi)(1 + i) pure imaginary (real part 0)?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

108

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Product: (2 − b) + (2 + b)i; real part 2 − b = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

True or false?

The product of two non-real complex numbers can be a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

109

QUESTION ID

S3-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example (1 + i)(1 − i) = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

For real c, $(c + i)^2$ is real when

A. $c = 0$

B. $c = 1$

C. $c = -1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

110

QUESTION ID

S3-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. c = 0

GRADING NOTE $(c + i)^2 = c^2 - 1 + 2ci$; imaginary part $2c = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q4**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

Show that (a + bi)(a − bi) is a real number for all real a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

111

QUESTION ID

S3-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$a^2 - abi + abi - b^2i^2 = a^2 + b^2$, which is real.

GRADING NOTE Any equivalent algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 2

Divide complex numbers using conjugates • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Write (3 + 4i) / (1 + 2i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

113

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

11/5 − (2/5)i

GRADING NOTE Form required: a + bi. Multiply by (1 − 2i)/(1 − 2i): (3 + 4i)(1 − 2i)/5 = (11 − 2i)/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

Write 10 / (3 − i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

114

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 + i

GRADING NOTE Form required: a + bi. 10(3 + i)/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

True or false?

To divide by 2 − 5i, multiply numerator and denominator by 2 − 5i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

115

QUESTION ID

S3-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the conjugate 2 + 5i; $(2 - 5i)^2$ is not real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

(2 + i) / i equals

A. −1 + 2i

B. 2 + i

C. 1 − 2i

D. 1 + 2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

116

QUESTION ID

S3-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 − 2i

GRADING NOTE (2 + i)(−i)/(i · −i) = (−2i + 1)/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

Explain why multiplying by the conjugate of the denominator always gives a real denominator.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

117

QUESTION ID

S3-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$(c + di)(c - di) = c^2 + d^2$, a real number (and positive unless $c = d = 0$).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSN-CN.C.7 • THE COMPLEX NUMBER SYSTEM

Solve quadratic equations with complex solutions

MAIN	Algebra II main question	**HSN-CN.C.7**
BACKWARD 1	Use the quadratic formula and the discriminant	**HSA-REI.B.4.b**
BACKWARD 2	Write square roots of negative numbers with i	**HSN-CN.A.1**
BACKWARD 3	Solve equations of the form $(x - h)^2 = k$	**HSA-REI.B.4.b**
FORWARD 1	Find a quadratic from one complex solution	**HSN-CN.C.7**
FORWARD 2	Classify solutions for a family of quadratics	**HSN-CN.C.7**

SET 4 • HSN-CN.C.7

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 49 = 0$ over the complex numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

120

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7i or x = −7i

GRADING NOTE Form required: both solutions, exact. $x^2 = -49$, $x = \pm 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-M-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

Solve $x^2 - 4x + 13 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

121

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2 + 3i or

x = 2 − 3i

GRADING NOTE Form required: both, a + bi. Discriminant 16 − 52 = −36; x = (4 ± 6i)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

Solve $2x^2 + 2x + 5 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

122

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = −1/2 + (3/2)i or
x = −1/2 − (3/2)i

GRADING NOTE Form required: both, a + bi. Discriminant 4 − 40 = −36; x = (−2 ± 6i)/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

Solve $(x + 3)^2 = -20$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

123

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

$x = -3 + 2\sqrt{5}\,i$ or
$x = -3 - 2\sqrt{5}\,i$

GRADING NOTE Form required: both, exact. $x + 3 = \pm\sqrt{-20} = \pm 2\sqrt{5}\,i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

Solve $3x^2 - 6x + 7 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

124

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 1 + (2\sqrt{3}/3)i$ or
$x = 1 - (2\sqrt{3}/3)i$

GRADING NOTE Form required: both, exact, a + bi. Discriminant $36 - 84 = -48 = (4\sqrt{3}\,i)^2$; $x = (6 \pm 4\sqrt{3}\,i)/6$. Check: product 7/3 and sum 2 agree with c/a and −b/a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 1

Use the quadratic formula and the discriminant • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Find the discriminant $b^2 - 4ac$ of $x^2 - 4x + 13 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

126

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−36

GRADING NOTE Any equivalent form. 16 − 52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

Find the discriminant of $2x^2 + 2x + 5 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

127

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−36

GRADING NOTE Any equivalent form. 4 − 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

True or false?

If the discriminant is negative, the quadratic equation has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

128

QUESTION ID

S4-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A negative discriminant gives two non-real solutions and no real ones.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In $3x^2 - 6x + 7 = 0$, the values of a, b and c are

A. 3, −6, −7

B. 3, 6, 7

C. −3, −6, 7

D. 3, −6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

129

QUESTION ID

S4-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3, −6, 7

GRADING NOTE Standard form $ax^2 + bx + c = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

Write the quadratic formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

130

QUESTION ID

S4-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 2

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−36) in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

132

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−48) in simplest form with i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

133

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4√3 i

GRADING NOTE Form required: simplest radical form. 48 = 16 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

√(−20) equals

A. 2√5 i

B. 4√5 i

C. −2√5

D. 10i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

134

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2√5 i

GRADING NOTE √20 = 2√5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{(-49)} = -7.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

135

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(-49)} = 7i.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

Write (−2 + √(−36)) / 4 in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

136

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2 + (3/2)i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 3

Solve equations of the form $(x - h)^2 = k$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 2)^2 = 9$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

138

QUESTION ID

S4-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = 5 or x = −1

GRADING NOTE Form required: both. $x - 2 = \pm 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

Solve $(x + 1)^2 = 5$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

139

QUESTION ID

S4-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = −1 + √5 or
x = −1 − √5

GRADING NOTE Form required: both, exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

True or false?

Solving $x^2 = 16$ gives only $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

140

QUESTION ID

S4-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Also $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

The solutions of $(x + 4)^2 = 25$ are

A. 5 and −5

B. −1 and 9

C. 1 and −9

D. 21 and −29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

141

QUESTION ID

S4-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 and −9

GRADING NOTE $x + 4 = \pm 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

Solve $x^2 = 50$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

142

QUESTION ID

S4-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 5\sqrt{2}$ or $x = -5\sqrt{2}$

GRADING NOTE Form required: both, exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B3-Q5**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 1

Find a quadratic from one complex solution • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

$3 - 2i$ is a solution of $x^2 + bx + c = 0$, where b and c are real. Find b and c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

144

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

b = −6, c = 13

GRADING NOTE Form required: both. With real coefficients the other solution is 3 + 2i; sum 6 = −b, product 9 + 4 = 13 = c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

$-1 + 4i$ is a solution of a quadratic equation $x^2 + bx + c = 0$ with real b and c. Write the equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

145

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x + 17 = 0$

GRADING NOTE Any equivalent equation. Sum −2, product 1 + 16 = 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

True or false?

If 2 + i is a solution of a quadratic with real coefficients, then 2 − i is also a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

146

QUESTION ID

S4-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Non-real solutions of real quadratics come in conjugate pairs (± in the formula).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

Which quadratic with real coefficients has 5i as a solution?

A. $x^2 + 25 = 0$

B. $x^2 + 5 = 0$

C. $x^2 - 25 = 0$

D. $x^2 - 10x + 25 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

147

QUESTION ID

S4-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 25 = 0$

GRADING NOTE $(5i)^2 + 25 = -25 + 25 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

Explain why $x^2 - (2 + i)x = 0$ has the solution $2 + i$ but not the solution $2 - i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

148

QUESTION ID

S4-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Its coefficient 2 + i is not real, so solutions need not come in conjugate pairs: the solutions are 0 and 2 + i.

GRADING NOTE Any equivalent explanation naming the non-real coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 2

Classify solutions for a family of quadratics • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

For which real values of k does $x^2 + kx + 9 = 0$ have two non-real solutions?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

150

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-6 < k < 6$

GRADING NOTE Form required: an inequality. Discriminant $k^2 - 36 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

For which real values of c does $x^2 - 4x + c = 0$ have non-real solutions? Give them for c = 8.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

151

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

c > 4; for c = 8, x = 2 ± 2i

GRADING NOTE Form required: both parts. Discriminant $16 - 4c < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

True or false?

$x^2 + kx - 1 = 0$ has non-real solutions for some real k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

152

QUESTION ID

S4-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its discriminant $k^2 + 4$ is always positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

$x^2 + 6x + m = 0$ has exactly one (repeated) real solution when m =

A. 3

B. 36

C. −9

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

153

QUESTION ID

S4-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9

GRADING NOTE $36 - 4m = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q4**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

For c > 4, write the solutions of $x^2 - 4x + c = 0$ in terms of c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

154

QUESTION ID

S4-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 2 \pm i\sqrt{(c - 4)}$

GRADING NOTE Form required: exact in terms of c. $(4 \pm \sqrt{(16 - 4c)})/2 = 2 \pm \sqrt{(4 - c)} = 2 \pm i\sqrt{(c - 4)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

Factor polynomials over the complex numbers

MAIN	Algebra II main question	**HSN-CN.C.8**
BACKWARD 1	Factor a difference of squares	**HSA-SSE.A.2**
BACKWARD 2	Multiply complex conjugates	**HSN-CN.A.2**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Factor cubic and quartic polynomials over the complex numbers	**HSN-CN.C.8**
FORWARD 2	Write polynomials with real coefficients from complex zeros	**HSN-CN.C.9**

SET 5 • HSN-CN.C.8

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

157

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 5i)(x - 5i)$

GRADING NOTE Any order of factors. $(x + 5i)(x - 5i) = x^2 - 25i^2 = x^2 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

Factor $x^2 + 7$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

158

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + i\sqrt{7})(x - i\sqrt{7})$

GRADING NOTE Any order. $x^2 - (i\sqrt{7})^2 = x^2 + 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

Factor $4x^2 + 9$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

159

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(2x + 3i)(2x - 3i)$

GRADING NOTE Any equivalent factorization (for example $4(x + 3i/2)(x - 3i/2)$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

Factor $x^4 - 16$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

160

QUESTION ID

S5-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 2)(x - 2i)(x + 2i)$

GRADING NOTE Any order. $x^4 - 16 = (x^2 - 4)(x^2 + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

Factor $x^2 - 6x + 13$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

161

QUESTION ID

S5-M-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - (3 + 2i))(x - (3 - 2i))$

GRADING NOTE Any equivalent form. Its zeros are $3 \pm 2i$ (discriminant -16). Check: sum 6, product $9 + 4 = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 1

Factor a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 49$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

163

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 7)(x − 7)

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

Factor $9x^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

164

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(3x + 2)(3x − 2)

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

$x^4 - 1$ equals

A. $(x - 1)^4$

B. $(x^2 - 1)^2$

C. $(x^2 - 1)(x^2 + 1)$

D. $(x^2 + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

165

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x^2 - 1)(x^2 + 1)$

GRADING NOTE $(x^2)^2 - 1^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

166

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

Factor $x^2 - 5$ using radicals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

167

QUESTION ID

S5-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + \sqrt{5})(x - \sqrt{5})$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 2

Multiply complex conjugates • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(x + 5i)(x - 5i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

169

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 25$

GRADING NOTE Any equivalent form. $-25i^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(2x + 3i)(2x - 3i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

170

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$4x^2 + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

True or false?

$(3i)^2 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

171

QUESTION ID

S5-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(3i)^2 = 9i^2 = -9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

$(x - (1 + i))(x - (1 - i))$ equals

A. $x^2 + 2x + 2$

B. $x^2 - 2x + 2$

C. $x^2 + 2$

D. $x^2 - 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

172

QUESTION ID

S5-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x^2 - 2x + 2$

GRADING NOTE Sum 2, product 1 + 1 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

Simplify $(i\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

173

QUESTION ID

S5-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q5**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 25 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

175

QUESTION ID

S5-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = ±5i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B3-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

Solve $x^2 - 6x + 13 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

176

QUESTION ID

S5-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 3 \pm 2i$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B3-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

True or false?

If r is a zero of p(x), then x − r is a factor of p(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

177

QUESTION ID

S5-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Factor theorem, also for complex r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

The zeros of $x^2 + 4$ are

A. 2i and −2i

B. 4i and −4i

C. −4

D. 2 and −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

178

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2i and −2i

GRADING NOTE $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x + 5 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

179

QUESTION ID

S5-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x = −1 ± 2i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 1

Factor cubic and quartic polynomials over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^3 + 8$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

181

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 2)(x - 1 - i\sqrt{3})(x - 1 + i\sqrt{3})$

GRADING NOTE Any order. $x^3 + 8 = (x + 2)(x^2 - 2x + 4)$, and $x^2 - 2x + 4$ has zeros $1 \pm i\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^3 + 9x$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

182

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

Factor $x^4 + 5x^2 + 4$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

183

QUESTION ID

S5-F1-Q3

TYPE

Short answer

CORRECT ANSWER

(x + i)(x − i)(x + 2i)(x − 2i)

GRADING NOTE Any order. $(x^2 + 1)(x^2 + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

True or false?

$x^3 - 1 = (x - 1)(x^2 + x + 1)$, and $x^2 + x + 1$ has two non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

184

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Discriminant $1 - 4 = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

Which is a linear factor of $x^3 - 27$ over the complex numbers?

A. $x + 3i$

B. $x + 3$

C. $x - 3i$

D. $x - (-3/2 + (3\sqrt{3}/2)i)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

185

QUESTION ID

S5-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D.

$x - (-3/2 + (3\sqrt{3}/2)i)$

GRADING NOTE $x^3 - 27 = (x - 3)(x^2 + 3x + 9)$; $x^2 + 3x + 9$ has zeros $-3/2 \pm (3\sqrt{3}/2)i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 2

Write polynomials with real coefficients from complex zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Write a polynomial with real coefficients, leading coefficient 1 and least degree, whose zeros include 2i and 1 − i.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

187

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 4)(x^2 - 2x + 2) = x^4 - 2x^3 + 6x^2 - 8x + 8$

GRADING NOTE Any equivalent form. Conjugates −2i and 1 + i must also be zeros, so degree 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

What is the least degree of a polynomial with real coefficients that has zeros 3, 4i and 2 + i?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

188

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. −4i and 2 − i are also zeros: 1 + 2 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

True or false?

A polynomial with real coefficients can have exactly one non-real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

189

QUESTION ID

S5-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Non-real zeros come in conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

A cubic with real coefficients has zeros 5 and −i. Its third zero is

A. −5

B. 5i

C. 1

D. i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

190

QUESTION ID

S5-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. i

GRADING NOTE Conjugate of −i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

Explain why a polynomial with real coefficients and degree 3 must have at least one real zero.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

191

QUESTION ID

S5-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It has 3 complex zeros with multiplicity; non-real zeros pair up, so they come two at a time, leaving at least one real zero.

GRADING NOTE Any equivalent explanation using the Fundamental Theorem of Algebra and conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

SET 6 • HSN-CN.C.9 • THE COMPLEX NUMBER SYSTEM

Use the Fundamental Theorem of Algebra

MAIN	Algebra II main question	**HSN-CN.C.9**
BACKWARD 1	Find the degree of a polynomial	**HSA-SSE.A.1.a**
BACKWARD 2	Find zeros from factored form	**HSA-SSE.B.3.a**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Relate the complex zeros to the real x-intercepts of a graph	**HSA-APR.B.3**
FORWARD 2	Prove existence statements about real zeros	**HSN-CN.C.9**

SET 6 • HSN-CN.C.9

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does $p(x) = 3x^5 - x^2 + 7$ have? Name the theorem you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

194

QUESTION ID
S6-M-Q1

TYPE
Short answer

CORRECT ANSWER
5, by the Fundamental Theorem of Algebra (degree 5)

GRADING NOTE Form required: the number and the theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

List every complex zero of $p(x) = (x - 2)^3(x^2 + 9)$ with its multiplicity, and check that the multiplicities add to the degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

195

QUESTION ID
S6-M-Q2

TYPE
Short answer

CORRECT ANSWER
2 (multiplicity 3), 3i (multiplicity 1), −3i (multiplicity 1);
3 + 1 + 1 = 5
= degree

GRADING NOTE Form required: all zeros with multiplicities and the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

Which argument shows that every quadratic $p(x) = ax^2 + bx + c$ (real a, b, c, $a \neq 0$) has exactly two complex zeros, counted with multiplicity?

A. $x^2 - 5x + 6 = (x - 2)(x - 3)$ has two zeros, so every quadratic has two zeros.

B. By the quadratic formula the zeros are $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$. If $b^2 - 4ac < 0$ the square root does not exist, so there are no zeros; otherwise there are two, so a quadratic never has more than two zeros.

C. The graph of every quadratic is a parabola, and a parabola that opens up or down always crosses the x-axis in exactly two places, so every quadratic $ax^2 + bx + c$ has exactly two zeros.

D. The quadratic formula $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$ gives every zero: two real zeros if $b^2 - 4ac > 0$, one zero of multiplicity 2 if $b^2 - 4ac = 0$, and two non-real conjugate zeros if $b^2 - 4ac < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

196

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The quadratic formula $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$ gives every zero: two real zeros if $b^2 - 4ac > 0$, one zero of multiplicity 2 if $b^2 - 4ac = 0$, and two non-real conjugate zeros if $b^2 - 4ac < 0$.

GRADING NOTE Every case of the discriminant gives two zeros counted with multiplicity (for $D < 0$, $\sqrt{D} = i\sqrt{(-D)}$ is a complex number). The others check one example, use a false graph fact ($x^2 + 1$ never crosses), or ignore complex zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

Find every complex zero of $p(x) = x^4 - 81$ and check that the count agrees with the Fundamental Theorem of Algebra.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

197

QUESTION ID

S6-M-Q4

TYPE

Short answer

CORRECT ANSWER

3, −3, 3i, −3i; four zeros for degree 4

GRADING NOTE Form required: all four and the check. $(x^2 - 9)(x^2 + 9)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

A student says $p(x) = x^3 + x + 1$ has no zeros because it does not factor with integers. Which statement is correct?

A. The student is right: a polynomial that does not factor over the integers has no zeros.

B. It has exactly 3 real zeros, because the Fundamental Theorem of Algebra says a polynomial of degree 3 has 3 zeros, and a polynomial with real coefficients has only real zeros.

C. It has 3 complex zeros counted with multiplicity, and at least one is real, because non-real zeros of a polynomial with real coefficients come in conjugate pairs.

D. It has 3 complex zeros counted with multiplicity, and all three could be non-real, because a polynomial that does not factor over the integers need not cross the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

198

QUESTION ID

S6-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. It has 3 complex zeros counted with multiplicity, and at least one is real, because non-real zeros of a polynomial with real coefficients come in conjugate pairs.

GRADING NOTE Degree 3 gives 3 zeros (Fundamental Theorem of Algebra); conjugate pairs make the number of non-real zeros even, so at least 1 is real. The others confuse "no integer factor" with "no zero", assume all zeros are real, or ignore conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 1

Find the degree of a polynomial • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

What is the degree of $3x^5 - x^2 + 7$?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

200

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

What is the degree of $(x - 2)^3(x^2 + 9)$ when multiplied out?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

201

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 3 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

The degree of $x(x + 1)(x^2 - 4)$ is

A. 1

B. 4

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

202

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE 1 + 1 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

True or false?

The degree of $4x^3 + 2x^4$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

203

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The highest power is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

What is the leading coefficient of $-2x^3 + 5x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

204

QUESTION ID

S6-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 2

Find zeros from factored form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the zeros of $(x - 2)(x + 5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

206

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 and −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B2-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

Find the zeros of $(x - 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

207

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1 (repeated three times)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

True or false?

A product is 0 only when one of its factors is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

208

QUESTION ID

S6-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Zero-product property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

The zeros of x(x + 4) are

A. 0 and 4

B. 0 and −4

C. −4 only

D. 4 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

209

QUESTION ID

S6-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0 and −4

GRADING NOTE x = 0 or x + 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the zeros of $(2x - 3)(x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

210

QUESTION ID

S6-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3/2 and −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 9 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

212

QUESTION ID

S6-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = ±3i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B3-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x + 2 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

213

QUESTION ID

S6-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = −1 ± i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

True or false?

$x^2 + 1 = 0$ has no complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

214

QUESTION ID

S6-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has ±i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

The solutions of $x^2 - 2x + 5 = 0$ are

A. $1 \pm 4i$

B. $-1 \pm 2i$

C. $1 \pm 2i$

D. $2 \pm i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

215

QUESTION ID

S6-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 ± 2i

GRADING NOTE Discriminant −16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = -81$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

216

QUESTION ID

S6-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x = ±9i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 1

Relate the complex zeros to the real x-intercepts of a graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A polynomial of degree 4 with real coefficients has a graph that crosses the x-axis at exactly two points and touches it nowhere else. Each crossing is a simple zero. How many non-real zeros does it have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

218

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 4 zeros in all, 2 real (the crossings), so 2 non-real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

The graph of $y = x^3 - 4x^2 + 5x - 2$ touches the x-axis at $x = 1$ and crosses it at $x = 2$. List all complex zeros with multiplicity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

219

QUESTION ID
S6-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1 (multiplicity 2) and 2 (multiplicity 1)

GRADING NOTE Form required: zeros with multiplicities. $(x - 1)^2(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 6 whose graph never meets the x-axis has 6 non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

220

QUESTION ID

S6-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All 6 zeros are non-real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

A degree-5 polynomial with real coefficients has exactly one real zero, of multiplicity 1. How many non-real zeros does it have?

A. 4

B. 5

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

221

QUESTION ID

S6-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE $5 - 1 = 4$ (two conjugate pairs).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

Can a degree-4 polynomial with real coefficients have exactly three real zeros, each of multiplicity 1? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

222

QUESTION ID

S6-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No: the fourth zero would be a single non-real zero, but non-real zeros come in conjugate pairs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 2

Prove existence statements about real zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Which argument proves that every polynomial of odd degree with real coefficients has at least one real zero?

A. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come one at a time, so there is an odd number of them, and an odd number of non-real zeros leaves at least one zero that is real.

B. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come in conjugate pairs, so there is an even number of them; odd minus even is odd, so at least one zero is real.

C. The Fundamental Theorem of Algebra says that every polynomial of degree $n \geq 1$ with real coefficients has at least one real zero; a polynomial of odd degree has degree at least 1, so it has at least one real zero as well.

D. $x^3 - 1$ has the real zero 1, so every polynomial of odd degree has a real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

224

QUESTION ID

S6-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come in conjugate pairs, so there is an even number of them; odd minus even is odd, so at least one zero is real.

GRADING NOTE Count zeros with the theorem and pair the non-real ones. The others check one example, misstate the theorem (it guarantees complex zeros, and $x^2 + 1$ has no real zero), or claim an odd number of non-real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

A degree-4 polynomial with real coefficients has the zeros 1 + i and 3. What are all of its possible zero lists?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

225

QUESTION ID
S6-F2-Q2

TYPE
Short answer

CORRECT ANSWER
1 + i, 1 − i, 3 and one more real zero r (r may be 3 again, giving 3 with multiplicity 2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

True or false?

A degree-4 polynomial with real coefficients must have at least one real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

226

QUESTION ID
S6-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $x^4 + 1$ has no real zeros; only odd degree guarantees a real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

A polynomial with real coefficients has degree 7. The possible numbers of real zeros (with multiplicity) are

A. 1, 3, 5, 7

B. 7 only

C. 0, 2, 4, 6

D. 0 to 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

227

QUESTION ID

S6-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1, 3, 5, 7

GRADING NOTE The number of real zeros is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

Explain why the Fundamental Theorem of Algebra says nothing about whether a polynomial's zeros can be found exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

228

QUESTION ID
S6-F2-Q5

TYPE
Short answer

CORRECT ANSWER
It guarantees that the zeros exist and counts them; it gives no method or formula for finding them.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a • SEEING STRUCTURE IN EXPRESSIONS

Interpret the parts of polynomial, rational, and exponential expressions in context

MAIN	Algebra II main question	**HSA-SSE.A.1.a**
BACKWARD 1	Name terms, factors, and coefficients	**HSA-SSE.A.1.a**
BACKWARD 2	Translate between a percent change and a multiplying factor	**7.RP.A.3**
BACKWARD 3	Interpret a quotient as an amount per unit	**6.RP.A.2**
FORWARD 1	Build an expression whose parts have given meanings	**HSA-SSE.A.1.a**
FORWARD 2	Rewrite an exponential expression to reveal a different rate	**HSF-IF.C.8.b**

SET 7 • HSA-SSE.A.1.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A town's population is modeled by $P(t) = 12{,}500(0.97)^t$, where t is the number of years after 2020. What do 12,500 and 0.97 represent, and what does $1 - 0.97$ represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

231

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

12,500 is the population in 2020; 0.97 is the yearly factor (each year the population is 97% of the year before); 1 − 0.97 = 0.03 means the population decreases by 3% per year.

GRADING NOTE Any equivalent interpretation of all three parts, with units (people, per year).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The average cost per item of making x items is C(x) = (4,000 + 6x) / x dollars. Interpret 4,000, 6x, and the division by x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

232

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

4,000 is a fixed cost in dollars; 6x is the variable cost ($6 per item for x items); dividing the total cost by x gives the cost per item.

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

An open box is made by cutting squares of side x cm from the corners of a 20 cm by 20 cm sheet. Its volume is $V(x) = x(20 - 2x)^2$ cm³. Interpret x, $20 - 2x$, and $(20 - 2x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

233

QUESTION ID

S7-M-Q3

TYPE

Short answer

CORRECT ANSWER

x is the height of the box; $20 - 2x$ is the side length of the square base (20 cm minus two cuts); $(20 - 2x)^2$ is the base area in cm².

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The amount of a medicine in the body t hours after a dose is $A(t) = 200(1/2)^{t/6}$ mg. Interpret 200, 1/2, and t/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

234

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

200 mg is the initial dose; the amount is halved repeatedly; t/6 counts 6-hour periods, so the amount halves every 6 hours (half-life 6 hours).

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

Maya keeps $1,500 in an account that earns 4% interest compounded once a year and also puts $200 per year in a jar (no interest). Her total after n years is $T(n) = 1500(1.04)^n + 200n$. Interpret $1500(1.04)^n$, 1.04, and 200n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

235

QUESTION ID
S7-M-Q5

TYPE
Short answer

CORRECT ANSWER
$1500(1.04)^n$ is the account balance after n years; 1.04 is the yearly growth factor (4% interest added each year); 200n is the cash in the jar after n years.

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 1

Name terms, factors, and coefficients • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

Name the terms of 4,000 + 6x.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q1/5

237

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4,000 and 6x

GRADING NOTE Form required: both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q1/5

What is the coefficient of x in 7x − 3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q2/5

238

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q2/5

In 5x(x − 2), which is a factor?

A. 5x − 2

B. x − 10

C. x − 2

D. −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q3/5

239

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x − 2

GRADING NOTE 5x(x − 2) is the product of 5, x and x − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q3/5

True or false?

In 3 + 2x, the number 3 is a factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q4/5

240

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 is a term (it is added).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the factors of $200(0.5)^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q5/5

241

QUESTION ID

S7-B1-Q5

TYPE

Short answer

CORRECT ANSWER

200 and $(0.5)^t$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 2

Translate between a percent change and a multiplying factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A price is multiplied by 0.97. By what percent does it change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q1/5

243

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

It decreases by 3%.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What factor multiplies an amount that increases by 4%?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q2/5

244

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1.04

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q2/5

Multiplying by 1.25 is the same as

A. increasing by 2.5%

B. increasing by 25%

C. increasing by 125%

D. decreasing by 25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q3/5

245

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. increasing by 25%

GRADING NOTE 1 + 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q3/5

True or false?

Multiplying by 0.8 decreases an amount by 80%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q4/5

246

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 20%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q4/5

Find 1 − 0.85 and explain what it means for an amount multiplied by 0.85.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q5/5

247

QUESTION ID

S7-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.15; the amount decreases by 15%.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 3

Interpret a quotient as an amount per unit • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A total cost of $600 is shared by 30 people. What does 600/30 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q1/5

249

QUESTION ID

S7-B3-Q1

TYPE

Short answer

CORRECT ANSWER

The cost per person, $20.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q1/5

A car travels 150 miles in 3 hours. What does 150/3 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q2/5

250

QUESTION ID

S7-B3-Q2

TYPE

Short answer

CORRECT ANSWER

The average speed, 50 miles per hour.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q2/5

If T is a total cost for n items, then T/n is

A. the profit

B. the number of items

C. the cost per item

D. the total cost

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q3/5

251

QUESTION ID

S7-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. the cost per item

GRADING NOTE Unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Dividing a total by the number of items gives an average per item.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q4/5

252

QUESTION ID

S7-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of a mean or unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q4/5

What does (4,000 + 6x)/x mean when x = 100?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q5/5

253

QUESTION ID

S7-B3-Q5

TYPE

Short answer

CORRECT ANSWER

The average cost per item when 100 items are made: $46.

GRADING NOTE Any equivalent answer. 4,600/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 1

Build an expression whose parts have given meanings • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Write an expression for the value of a $24,000 car that loses 15% of its value each year, t years after purchase.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q1/5

255

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$24{,}000(0.85)^t$

GRADING NOTE Any equivalent expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges a $90 joining fee and $35 per month. Write an expression for the average cost per month over m months.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q2/5

256

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(90 + 35m)/m

GRADING NOTE Any equivalent expression (for example 35 + 90/m).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q2/5

True or false?

An expression for a quantity that triples every 5 years, starting at 40, is $40(3)^{5t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q3/5

257

QUESTION ID
S7-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The exponent counts 5-year periods: $40(3)^{t/5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A bacteria count starts at 500 and doubles every 3 hours. Which expression gives the count after h hours?

A. $2(500)^{h/3}$

B. $500(2)^{3h}$

C. $500(3)^{h/2}$

D. $500(2)^{h/3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q4/5

258

QUESTION ID

S7-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $500(2)^{h/3}$

GRADING NOTE h/3 doubling periods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q4/5

A gym charges a $90 joining fee and $35 per month, so the average cost per month over m months is 35 + 90/m dollars. Explain what the part 90/m means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q5/5

259

QUESTION ID

S7-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The joining fee spread over m months: the part of the monthly average that comes from the fee, which shrinks as m grows.

GRADING NOTE Any equivalent explanation (the gym charges $90 to join and $35 per month).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 2

Rewrite an exponential expression to reveal a different rate • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Rewrite $12{,}500(0.97)^t$ (t in years) to show the monthly factor, rounded to 5 decimal places.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q1/5

261

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$12{,}500(0.99746)^{12t}$

GRADING NOTE Rounding to 5 decimal places required for the factor. $0.97^{1/12} \approx 0.997465$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$A(t) = 200(1/2)^{t/6}$. Write A as $200 \cdot b^t$ and give b to 4 decimal places. What is the hourly percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q2/5

262

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$b = (1/2)^{1/6} \approx 0.8909$; about 10.91% decrease per hour

GRADING NOTE Rounding to 4 decimal places for b; percent to 0.01.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q2/5

True or false?

$1.04^{n} = (1.04^{1/12})^{12n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q3/5

263

QUESTION ID

S7-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Power of a power: (1/12)(12n) = n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q3/5

Which expression shows that $500(2)^{t/3}$ grows by about 26% per time unit?

A. $1.26(500)^t$

B. $500(1.26)^{3t}$

C. $500(1.26)^t$

D. $500(2.26)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q4/5

264

QUESTION ID

S7-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $500(1.26)^t$

GRADING NOTE $2^{1/3} \approx 1.26$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q4/5

Explain why rewriting does not change the values of the function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q5/5

265

QUESTION ID

S7-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The new form is equivalent by the properties of exponents, so it gives the same output for every input; it only shows another rate.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b • SEEING STRUCTURE IN EXPRESSIONS

Interpret complicated expressions by viewing parts as single entities

MAIN	Algebra II main question	**HSA-SSE.A.1.b**
BACKWARD 1	Squares are never negative	**8.EE.A.2**
BACKWARD 2	Powers of numbers greater than 1 and between 0 and 1	**8.EE.A.1**
BACKWARD 3	A fraction gets smaller as its denominator grows	**3.NF.A.3.d**
FORWARD 1	Find the set of all values of an expression on a domain	**HSA-SSE.A.1.b**
FORWARD 2	Use single-entity reasoning to find horizontal asymptotes	**HSF-IF.C.7.d**

SET 8 • HSA-SSE.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

$P(r) = 5000(1 + r)^{10}$ with $r > 0$ is the value of an investment. Without calculating, explain why $P(r) > 5000$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

268

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

$1 + r > 1$, so $(1 + r)^{10} > 1$, and 5000 times a number greater than 1 is greater than 5000.

GRADING NOTE Any equivalent explanation that treats $(1 + r)^{10}$ as one quantity greater than 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

The height of a ball is $h(t) = 80 - 16(t - 2)^2$ feet after t seconds. Without expanding, find the greatest height and when it occurs, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

269

QUESTION ID
S8-M-Q2

TYPE
Short answer

CORRECT ANSWER
80 ft at t = 2: $(t - 2)^2 \geq 0$, so $16(t - 2)^2 \geq 0$ and $h(t) \leq 80$, with equality only when $t = 2$.

GRADING NOTE Form required: the height, the time, and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

$C(x) = 10 + 40/(x + 1)$ is a cost in dollars for $x \geq 0$. Explain why $C(x)$ is always more than 10 and never more than 50.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

270

QUESTION ID

S8-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x + 1 \geq 1$, so $40/(x + 1)$ is positive and at most 40; adding 10 gives more than 10 and at most 50 (equal to 50 at $x = 0$).

GRADING NOTE Any equivalent explanation treating $40/(x + 1)$ as one positive quantity at most 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

The temperature of a drink is $T(t) = 70 - 30(0.8)^t$ °F after t minutes. Explain why T(t) is always below 70 and gets closer to 70 as t grows.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

271

QUESTION ID

S8-M-Q4

TYPE

Short answer

CORRECT ANSWER

$30(0.8)^t$ is positive, so T is below 70; $(0.8)^t$ shrinks toward 0 as t grows, so the subtracted part shrinks and T approaches 70.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

$f(x) = 3/(x^2 + 1)$. Find the greatest value of $f(x)$ and explain why no input gives a greater value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

272

QUESTION ID

S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

3, at $x = 0$: $x^2 + 1 \geq 1$, so $3/(x^2 + 1) \leq 3$, with equality only when $x = 0$.

GRADING NOTE Form required: the value and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 1

Squares are never negative • Grade 8 • Expressions & Equations • 8.EE.A.2

True or false?

$(t - 2)^2 \geq 0$ for every real t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q1/5

274

QUESTION ID

S8-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A square of a real number is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the smallest value of $(x + 5)^2$, and for which x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q2/5

275

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0, when x = −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q2/5

Which expression can be negative for some real x?

A. $x^2 + 1$

B. $x - 1$

C. x^2

D. $(x - 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q3/5

276

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x − 1

GRADING NOTE $x - 1 < 0$ for $x < 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q3/5

What is the smallest value of $x^2 + 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q4/5

277

QUESTION ID

S8-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. At x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q4/5

True or false?

$-(x - 3)^2$ is never positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q5/5

278

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the opposite of a nonnegative number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 2

Powers of numbers greater than 1 and between 0 and 1 • Grade 8 • Expressions & Equations • 8.EE.A.1

Is 1.05^{10} greater than or less than 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q1/5

280

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Greater than 1

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q1/5

Is 0.8^5 greater than or less than 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q2/5

281

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Less than 1 (about 0.33)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As n increases, 0.5^n

A. becomes negative

B. stays 0.5

C. gets closer to 0

D. grows without bound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q3/5

282

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. gets closer to 0

GRADING NOTE Repeated halving.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q3/5

True or false?

A negative number raised to any whole-number power is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q4/5

283

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-2)^2 = 4$ is positive; even powers of negatives are positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Order from least to greatest: 0.9^2, 0.9^5, 0.9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q5/5

284

QUESTION ID

S8-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.9^5, 0.9^2, 0.9

GRADING NOTE Any equivalent listing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 3

A fraction gets smaller as its denominator grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Which is greater, 40/2 or 40/5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q1/5

286

QUESTION ID

S8-B3-Q1

TYPE

Short answer

CORRECT ANSWER

40/2

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q1/5

True or false?

If $x \geq 0$, then $40/(x + 1) \leq 40$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q2/5

287

QUESTION ID

S8-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The denominator is at least 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As x grows, 6/(x + 3)

A. increases

B. becomes negative

C. decreases toward 0

D. stays 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q3/5

288

QUESTION ID

S8-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. decreases toward 0

GRADING NOTE The denominator grows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q3/5

What is the greatest value of 12/(x + 4) for x ≥ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q4/5

289

QUESTION ID

S8-B3-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. At x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q4/5

True or false?

8/10 is greater than 8/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q5/5

290

QUESTION ID

S8-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Same numerator, larger denominator gives a smaller fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 1

Find the set of all values of an expression on a domain • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

Find every value that $h(t) = 80 - 16(t - 2)^2$ takes for $0 \le t \le 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q1/5

292

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$16 \le h \le 80$

GRADING NOTE Form required: an inequality or interval. $(t - 2)^2$ goes from 0 to 4, so $16(t - 2)^2$ from 0 to 64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find every value of $f(x) = 3/(x^2 + 1)$ for real x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q2/5

293

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$0 < f(x) \leq 3$

GRADING NOTE Form required: an inequality or interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q2/5

True or false?

For $x \geq 0$, $C(x) = 10 + 40/(x + 1)$ takes every value in $10 < C \leq 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q3/5

294

QUESTION ID

S8-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 40/(x + 1) takes every value in (0, 40].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q3/5

For $t \geq 0$, which set contains every value of $70 - 30(0.8)^t$?

A. $T \geq 70$

B. $40 \leq T < 70$

C. $40 < T \leq 70$

D. $0 < T < 70$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q4/5

295

QUESTION ID

S8-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $40 \leq T < 70$

GRADING NOTE $30(0.8)^t$ goes from 30 (at $t = 0$) toward 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q4/5

Find every value of $g(x) = 5 - 2/(x^2 + 4)$ for real x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q5/5

296

QUESTION ID

S8-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$4.5 \le g(x) < 5$

GRADING NOTE Form required: an inequality or interval. $2/(x^2 + 4)$ is in $(0, 0.5]$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 2

Use single-entity reasoning to find horizontal asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Explain why y = 10 + 40/(x + 1) has the horizontal asymptote y = 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q1/5

298

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

As x grows without bound, 40/(x + 1) gets closer to 0, so y gets closer to 10.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q1/5

Rewrite (2x + 7)/(x + 3) as 2 + 1/(x + 3) and state its horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q2/5

299

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2

GRADING NOTE Form required: the equation y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q2/5

True or false?

$y = 3/(x^2 + 1)$ has the horizontal asymptote $y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q3/5

300

QUESTION ID

S8-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3/(x^2 + 1)$ approaches 0 as $|x|$ grows; the asymptote is $y = 0$ (3 is the maximum).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q3/5

The horizontal asymptote of $y = 5 - 2/(x - 4)$ is

A. $y = -2$

B. $y = 5$

C. $y = 4$

D. $y = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q4/5

301

QUESTION ID
S8-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. y = 5

GRADING NOTE $-2/(x - 4)$ approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q4/5

Does the graph of y = 10 + 40/(x + 1) ever reach its horizontal asymptote? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q5/5

302

QUESTION ID

S8-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No: 40/(x + 1) is never 0, so y is never exactly 10.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q5/5

Factor higher-degree expressions by recognizing differences of squares and sums or differences of cubes

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Take out a greatest common factor	**6.EE.A.3**
BACKWARD 2	Recognize perfect squares and perfect cubes	**8.EE.A.2**
BACKWARD 3	Multiply to check a factoring pattern	**HSA-APR.A.1**
FORWARD 1	Factor expressions whose parts are themselves expressions	**HSA-SSE.A.2**
FORWARD 2	Complete the factorization over the complex numbers	**HSN-CN.C.8**

SET 9 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 81$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q1/5

305

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 9)(x + 3)(x - 3)$

GRADING NOTE Form required: factored completely over the integers. $(x^2)^2 - 9^2 = (x^2 - 9)(x^2 + 9)$. $x^2 + 9$ does not factor over the integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q1/5

Factor $8x^3 + 27$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q2/5

306

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 3)(4x^2 - 6x + 9)$

GRADING NOTE Form required: factored completely. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ with $a = 2x$, $b = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q2/5

Factor $x^6 - 64$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q3/5

307

QUESTION ID

S9-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x^2 + 2x + 4)(x + 2)(x^2 - 2x + 4)$

GRADING NOTE Form required: factored completely, any order. $(x^3 - 8)(x^3 + 8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q3/5

Factor $16x^4 - y^4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q4/5

308

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(4x^2 + y^2)(2x + y)(2x - y)$

GRADING NOTE Form required: factored completely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q4/5

Factor $2x^5 - 32x$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q5/5

309

QUESTION ID

S9-M-Q5

TYPE

Short answer

CORRECT ANSWER

$2x(x^2 + 4)(x + 2)(x - 2)$

GRADING NOTE Form required: factored completely. Common factor 2x first: $2x(x^4 - 16)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 1

Take out a greatest common factor • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Factor 6x + 18.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q1/5

311

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6(x + 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q1/5

Factor $2x^5 - 32x$ by taking out the greatest common factor only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q2/5

312

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$2x(x^4 - 16)$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q2/5

The greatest common factor of $12x^3$ and $18x$ is

A. $6x^3$

B. $6x$

C. $3x$

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q3/5

313

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6x

GRADING NOTE 6 and x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$3x^2 + 9x = 3x(x + 9)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q4/5

314

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3x(x + 9) = 3x^2 + 27x$; the correct factorization is $3x(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q4/5

Factor $5y^2 - 20y$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q5/5

315

QUESTION ID

S9-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5y(y − 4)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 2

Recognize perfect squares and perfect cubes • Grade 8 • Expressions & Equations • 8.EE.A.2

Write $16x^4$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q1/5

317

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(4x^2)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q1/5

Write $8x^3$ as a cube.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q2/5

318

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x)^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q2/5

Which is a perfect cube?

A. 32

B. 16

C. 36

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q3/5

319

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 64

GRADING NOTE $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q3/5

True or false?

x^6 is both a perfect square and a perfect cube.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q4/5

320

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x^3)^2$ and $(x^2)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the cube root of 27.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q5/5

321

QUESTION ID

S9-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 3

Multiply to check a factoring pattern • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(a + b)(a^2 - ab + b^2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q1/5

323

QUESTION ID

S9-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$a^3 + b^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x - 2)(x^2 + 2x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q2/5

324

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 - 8$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q2/5

True or false?

$(x + 3)(x^2 - 6x + 9) = x^3 + 27.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q3/5

325

QUESTION ID

S9-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The middle term should be $-3x$: $(x + 3)(x^2 - 3x + 9) = x^3 + 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x^2 - 9)(x^2 + 9)$ equals

A. $x^4 - 18x^2 - 81$

B. $x^4 + 81$

C. $(x - 3)^4$

D. $x^4 - 81$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q4/5

326

QUESTION ID

S9-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x^4 - 81$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q4/5

Multiply $(2x + 1)(4x^2 - 2x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q5/5

327

QUESTION ID

S9-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$8x^3 + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 1

Factor expressions whose parts are themselves expressions • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $(x + 1)^2 - 9$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q1/5

329

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 4)$

GRADING NOTE Form required: factored completely. $((x + 1) - 3)((x + 1) + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $(2x - 1)^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q2/5

330

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 1)(4x^2 - 8x + 7)$

GRADING NOTE Form required: factored completely. a = 2x − 1, b = 2: $(a + b)(a^2 - 2a + 4) = (2x + 1)((2x - 1)^2 - 2(2x - 1) + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q2/5

True or false?

$a^6 - b^6$ can be factored first as a difference of squares or first as a difference of cubes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q3/5

331

QUESTION ID

S9-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(a^3)^2 - (b^3)^2$ or $(a^2)^3 - (b^2)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q3/5

$(x - y)^2 - (x + y)^2$ equals

A. 0

B. $4xy$

C. $-4xy$

D. $-2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q4/5

332

QUESTION ID

S9-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-4xy$

GRADING NOTE $((x - y) - (x + y))((x - y) + (x + y)) = (-2y)(2x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q4/5

Factor $x^2(x + 2) - 4(x + 2)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q5/5

333

QUESTION ID

S9-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + 2)^2(x - 2)$

GRADING NOTE Form required: factored completely. $(x + 2)(x^2 - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 2

Complete the factorization over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^4 - 81$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q1/5

335

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(x - 3)(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $16x^4 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q2/5

336

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 1)(2x - 1)(2x + i)(2x - i)$

GRADING NOTE Any equivalent factorization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q2/5

True or false?

Over the complex numbers, $x^3 + 8$ has only one linear factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q3/5

337

QUESTION ID

S9-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has three: $(x + 2)$ and two from $x^2 - 2x + 4 = (x - 1 - i\sqrt{3})(x - 1 + i\sqrt{3})$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q3/5

Which polynomial is fully factored over the integers but not over the complex numbers?

A. $x^2 + 9$

B. $x - 9$

C. $x^2 - 9$

D. $(x - 3)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q4/5

338

QUESTION ID

S9-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 9$

GRADING NOTE $x^2 + 9 = (x + 3i)(x - 3i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q4/5

$2x^5 - 32x = 2x(x^2 + 4)(x + 2)(x - 2)$ over the integers. Complete the factorization over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q5/5

339

QUESTION ID

S9-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2x(x + 2i)(x - 2i)(x + 2)(x - 2)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q5/5

SET 10 • HSA-SSE.A.2 • SEEING STRUCTURE IN EXPRESSIONS

Rewrite expressions in quadratic form and factor

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Factor quadratic trinomials	**HSA-SSE.A.2**
BACKWARD 2	Write powers as squares of other powers	**8.EE.A.1**
BACKWARD 3	Treat a repeated part as a single variable	**HSA-SSE.A.1.b**
FORWARD 1	Solve equations in quadratic form over the complex numbers	**HSN-CN.C.7**
FORWARD 2	Solve exponential equations in quadratic form	**HSF-LE.A.4**

SET 10 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 5x^2 + 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q1/5

342

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1)(x - 2)(x + 2)$

GRADING NOTE Form required: factored completely. With $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q1/5

Factor $x^4 + 3x^2 - 10$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q2/5

343

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 5)(x^2 - 2)$

GRADING NOTE Form required: factored completely over the integers ($x^2 - 2$ does not factor over the integers). $u^2 + 3u - 10 = (u + 5)(u - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q2/5

Factor $9^x - 4 \cdot 3^x + 3$ by writing it in quadratic form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q3/5

344

QUESTION ID

S10-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(3^x - 1)(3^x - 3)$

GRADING NOTE Form required: a product of two factors. $9^x = (3^x)^2$, so with $u = 3^x$: $u^2 - 4u + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q3/5

Factor $x^6 - 9x^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q4/5

345

QUESTION ID

S10-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x^2 + x + 1)(x - 2)(x^2 + 2x + 4)$

GRADING NOTE Form required: factored completely. $u = x^3$: $(u - 1)(u - 8)$; then difference of cubes twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q4/5

Factor $2x^4 - 7x^2 - 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q5/5

346

QUESTION ID

S10-M-Q5

TYPE

Short answer

CORRECT ANSWER

$(2x^2 + 1)(x - 2)(x + 2)$

GRADING NOTE Form required: factored completely. $2u^2 - 7u - 4 = (2u + 1)(u - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 1

Factor quadratic trinomials • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $u^2 - 5u + 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q1/5

348

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(u − 1)(u − 4)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B1-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q1/5

Factor $u^2 + 3u - 10$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q2/5

349

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(u + 5)(u − 2)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B1-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q2/5

$2u^2 - 7u - 4$ equals

A. $(2u + 1)(u - 4)$

B. $(u - 2)(2u + 2)$

C. $(2u + 4)(u - 1)$

D. $(2u - 1)(u + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q3/5

350

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2u + 1)(u − 4)

GRADING NOTE Check: $2u^2 - 8u + u - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$u^2 - 9u + 8 = (u + 1)(u - 8)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q4/5

351

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(u + 1)(u - 8) = u^2 - 7u - 8$; the correct factorization is $(u - 1)(u - 8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $u^2 - 4u + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q5/5

352

QUESTION ID

S10-B1-Q5

TYPE

Short answer

CORRECT ANSWER

(u − 1)(u − 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 2

Write powers as squares of other powers • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write x^4 as a power of x^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

354

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

Write 9^x as a power of 3^x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

355

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(3^x)^2$

GRADING NOTE Any equivalent form. $9^x = (3^2)^x = (3^x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

x^6 equals

A. $(x^4)^2$

B. $(x^3)^2$

C. $x^3 \cdot x^3 \cdot x^3$

D. $(x^2)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

356

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x^3)^2$

GRADING NOTE $3 \cdot 2 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$4^x = (2^x)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

357

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^x = 2^{2x}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

Write 25^x as a power of 5^x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

358

QUESTION ID

S10-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(5^x)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 3

Treat a repeated part as a single variable • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

In $x^4 - 5x^2 + 4$, what single variable u turns the expression into $u^2 - 5u + 4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q1/5

360

QUESTION ID

S10-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$u = x^2$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $2(x - 1)^2 + 3(x - 1) - 5$ using $u = x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q2/5

361

QUESTION ID

S10-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$2u^2 + 3u - 5$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q2/5

True or false?

In $4^x - 2^x - 6$, the substitution $u = 2^x$ gives $u^2 - u - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q3/5

362

QUESTION ID

S10-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^x = (2^x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q3/5

Which substitution writes $x^6 - 9x^3 + 8$ as a quadratic?

A. $u = x^2$

B. $u = x^3$

C. $u = x^6$

D. $u = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q4/5

363

QUESTION ID

S10-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $u = x^3$

GRADING NOTE $x^6 = (x^3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q4/5

After factoring $(u - 1)(u - 8)$ with $u = x^3$, write the result in terms of x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q5/5

364

QUESTION ID

S10-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$(x^3 - 1)(x^3 - 8)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 1

Solve equations in quadratic form over the complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^4 + 3x^2 - 4 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q1/5

366

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 1, -1, 2i, -2i$

GRADING NOTE Form required: all four. $x^2 = 1$ or $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q1/5

Solve $x^4 - 5x^2 + 4 = 0$ over the complex numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q2/5

367

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = ±1, ±2

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q2/5

True or false?

$x^4 + 10x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q3/5

368

QUESTION ID

S10-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x^2 + 1)(x^2 + 9) = 0$ has only the solutions $\pm i$ and $\pm 3i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q3/5

How many complex solutions, with multiplicity, does $x^4 - 13x^2 + 36 = 0$ have?

A. 1

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q4/5

369

QUESTION ID

S10-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE ±2 and ±3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q4/5

Solve $x^4 + x^2 - 12 = 0$ over the complex numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q5/5

370

QUESTION ID

S10-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x = \pm\sqrt{3}, \pm 2i$

GRADING NOTE Form required: all four, exact. $(x^2 + 4)(x^2 - 3) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 2

Solve exponential equations in quadratic form • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $9^x - 5 \cdot 3^x + 6 = 0$. Give exact solutions.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q1/5

372

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \log_3 2$ or $x = 1$

GRADING NOTE Form required: both, exact. $3^x = 2$ or $3^x = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q1**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q1/5

Solve $4^x - 6 \cdot 2^x + 8 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q2/5

373

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 1 or x = 2

GRADING NOTE Form required: both. $2^x = 2$ or $2^x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q2/5

True or false?

$4^x + 2^x - 2 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q3/5

374

QUESTION ID

S10-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^x = 1$ gives $x = 0$; $2^x = -2$ is impossible since $2^x > 0$. One solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solutions of $25^x - 26 \cdot 5^x + 25 = 0$ are

A. 0 and 2

B. 1 and 25

C. 0 and 1

D. 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q4/5

375

QUESTION ID
S10-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 0 and 2

GRADING NOTE $5^x = 1$ or $5^x = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q4/5

Solve $e^{2t} - 7e^{t} + 10 = 0$. Give exact solutions and decimal values to 3 decimal places.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q5/5

376

QUESTION ID

S10-F2-Q5

TYPE

Short answer

CORRECT ANSWER

t = ln 2 ≈ 0.693 or
t = ln 5 ≈ 1.609

GRADING NOTE Form required: exact and 3 decimal places. $e^{t} = 2$ or $e^{t} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4 • SEEING STRUCTURE IN EXPRESSIONS

Derive the formula for the sum of a finite geometric series

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Write terms of a geometric sequence	**HSF-LE.A.2**
BACKWARD 2	Distribute a factor over a sum	**6.EE.A.3**
BACKWARD 3	Use the product rule for exponents	**8.EE.A.1**
FORWARD 1	Prove the sum formula from a polynomial identity	**HSA-APR.C.4**
FORWARD 2	Describe how partial sums behave when the ratio is between -1 a...	**HSA-SSE.B.4**

SET 11 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

$S_n = a + ar + ar^2 + \ldots + ar^{n-1}$ with $r \neq 1$. In the derivation of $S_n = a(1 - r^n)/(1 - r)$, which equation results from subtracting rS_n from S_n?

A. $S_n - rS_n = a - ar^n$

B. $S_n - rS_n = ar - ar^n$

C. $S_n - rS_n = a - ar$

D. $S_n - rS_n = a - ar^{n-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q1/5

379

QUESTION ID

S11-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $S_n - rS_n = a - ar^n$

GRADING NOTE $rS_n = ar + ar^2 + \ldots + ar^n$, so every middle term cancels and $S_n - rS_n = a - ar^n$. Then $(1 - r)S_n = a(1 - r^n)$, and dividing by $1 - r$ (allowed because $r \neq 1$) gives the formula. The others use the wrong last term of rS_n, cancel too much, or lose the first term a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q1/5

$S = 3 + 6 + 12 + \ldots + 3 \cdot 2^9$. With the multiply-and-subtract method, $2S - S$ equals which expression, and what is S?

A. $3 \cdot 2^9 - 3$, so S = 1533

B. $3 \cdot 2^{10} + 3$, so S = 3075

C. $3 \cdot 2^{10} - 6$, so S = 3066

D. $3 \cdot 2^{10} - 3$, so S = 3069

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q2/5

380

QUESTION ID

S11-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $3 \cdot 2^{10} - 3$, so S = 3069

GRADING NOTE $2S = 6 + 12 + \ldots + 3 \cdot 2^{10}$; subtracting S leaves the last term of 2S minus the first term of S: $3 \cdot 2^{10} - 3 = 3072 - 3 = 3069$. The others use the wrong last term, subtract the wrong first term, or add it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q2/5

Why does the derivation of $S_n = a(1 - r^n)/(1 - r)$ fail when $r = 1$, and what is S_n then?

A. When $r = 1$ the terms a, ar, ar^2, ... grow without bound, so S_n does not exist and no formula for it is possible.

B. Its last step divides by $1 - r$, which is 0 when $r = 1$. Every term then equals a, so $S_n = na$.

C. rS_n cannot be formed when $r = 1$, so $S_n = a$.

D. The formula gives 0/0, which means $S_n = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q3/5

381

QUESTION ID

S11-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Its last step divides by $1 - r$, which is 0 when $r = 1$. Every term then equals a, so $S_n = na$.

GRADING NOTE Division by $1 - r = 0$ is not allowed; with $r = 1$ the series is $a + a + \ldots + a$ (n terms) $= na$. The others treat 0/0 as 0, misread the method, or confuse $r = 1$ with $|r| > 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q3/5

$T = 1 + x + x^2 + \ldots + x^n$ ($n + 1$ terms, $x \neq 1$).
Multiplying by x and subtracting T gives which formula?
(Check: $x = 2$, $n = 3$ gives $1 + 2 + 4 + 8 = 15$.)

A. $T = (x^n - 1)/(x - 1)$

B. $T = (x^{n+1} - x)/(x - 1)$

C. $T = (x^{n+1} - 1)/(x - 1)$

D. $T = (x^{n+1} - 1)/(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q4/5

382

QUESTION ID

S11-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C.

$T = (x^{n+1} - 1)/(x - 1)$

GRADING NOTE $xT = x + x^2 + \ldots + x^{n+1}$, so $xT - T = x^{n+1} - 1$ and $T = (x^{n+1} - 1)/(x - 1)$. Check: $(16 - 1)/1 = 15$. The others give 7, 5, and 14 for the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q4/5

$T = 100 + 100(1.05) + 100(1.05)^2 + \ldots + 100(1.05)^{19}$ (20 yearly amounts). Using the multiply-and-subtract method with $1.05T - T$, which equation and total are correct? A calculator may be used.

A. $0.05T = 100(1.05^{19} - 1)$, so $T \approx \$3{,}053.90$

B. $1.05T = 100(1.05^{20} - 1)$, so $T \approx \$157.46$

C. $0.05T = 100(1.05^{21} - 1)$, so $T \approx \$3{,}571.93$

D. $0.05T = 100(1.05^{20} - 1)$, so $T \approx \$3{,}306.60$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q5/5

383

QUESTION ID

S11-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $0.05T = 100(1.05^{20} - 1)$, so $T \approx \$3{,}306.60$

GRADING NOTE $1.05T = 100(1.05) + \ldots + 100(1.05)^{20}$, so $1.05T - T = 100(1.05)^{20} - 100$ and $T = 100(1.05^{20} - 1)/0.05 \approx 3306.60$. The others use the wrong last power (19 or 21) or forget that $1.05T - T = 0.05T$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 1

Write terms of a geometric sequence • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the first four terms of the geometric sequence with first term 3 and ratio 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q1/5

385

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3, 6, 12, 24

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the 10th term of 3, 6, 12, ...?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q2/5

386

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$3 \cdot 2^9 = 1536$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q2/5

The nth term of a geometric sequence with first term a and ratio r is

A. $a + (n - 1)r$

B. $a \cdot n \cdot r$

C. ar^n

D. ar^{n-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q3/5

387

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ar^{n-1}

GRADING NOTE The first term is ar^0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q3/5

True or false?

100, 105, 110.25, ... is arithmetic with common difference 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q4/5

388

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences are 5 and 5.25; the ratio 1.05 is constant, so it is geometric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the ratio of 81, 27, 9, 3, ...

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q5/5

389

QUESTION ID

S11-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 2

Distribute a factor over a sum • Grade 6 • Expressions & Equations • 6.EE.A.3

Distribute: 2(3 + 6 + 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q1/5

391

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 + 12 + 24

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distribute: $r(a + ar + ar^2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q2/5

392

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$ar + ar^2 + ar^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q2/5

True or false?

$x(1 + x + x^2) = x + x^2 + x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q3/5

393

QUESTION ID

S11-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distributive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q3/5

1.05(100 + 105) equals

A. 105 + 105

B. 100 + 110.25

C. 105 + 110.25

D. 205

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q4/5

394

QUESTION ID

S11-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 105 + 110.25

GRADING NOTE Each term is multiplied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q4/5

Subtract: (6 + 12 + 24 + 48) − (3 + 6 + 12 + 24).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q5/5

395

QUESTION ID

S11-B2-Q5

TYPE

Short answer

CORRECT ANSWER

45 (= 48 − 3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 3

Use the product rule for exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $r \cdot r^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q1/5

397

QUESTION ID

S11-B3-Q1

TYPE

Short answer

CORRECT ANSWER

r^5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $ar^{n-1} \cdot r$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q2/5

398

QUESTION ID

S11-B3-Q2

TYPE

Short answer

CORRECT ANSWER

ar^n

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$2 \cdot 2^9$ equals

A. 2^{10}

B. 4^{10}

C. 4^9

D. 2^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q3/5

399

QUESTION ID

S11-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2^{10}

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q3/5

True or false?

$x \cdot x^n = x^{2n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q4/5

400

QUESTION ID

S11-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Add exponents: $x^1 \cdot x^n = x^{n+1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q4/5

Evaluate 2^{10}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q5/5

401

QUESTION ID

S11-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1024

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 1

Prove the sum formula from a polynomial identity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Which argument proves the identity
$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \ldots + x + 1)$?

A. Divide both sides by $x - 1$ to get $(x^n - 1)/(x - 1) = x^{n-1} + \ldots + x + 1$; this is the geometric sum formula, so the identity is true.

B. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and -1, leaving $x^n - 1$.

C. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and 1, leaving $x^n + 1$.

D. For $x = 2$ and $n = 3$: $7 = 1 \cdot 7$, so the identity holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q1/5

403

QUESTION ID
S11-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and -1, leaving $x^n - 1$.

GRADING NOTE Distribute and cancel (telescoping). The others check one example, assume what is to be proved (and divide by $x - 1$, which can be 0), or make a sign error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q1/5

Use the identity $x^n - 1 = (x - 1)(x^{n-1} + \ldots + 1)$ to write $1 + 3 + 9 + \ldots + 3^5$ as a quotient, and evaluate it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q2/5

404

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(3^6 - 1)/(3 - 1)$
$= 728/2 = 364$

GRADING NOTE Form required: the quotient and the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q2/5

True or false?

The identity shows that $x - 1$ is a factor of $x^n - 1$ for every whole number $n \geq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q3/5

405

QUESTION ID

S11-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It writes $x^n - 1$ as $(x - 1)$ times a polynomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q3/5

Which equals $1 + 2 + 4 + \ldots + 2^9$?

A. 2^{10}

B. $2^{10} - 1$

C. $2^9 - 1$

D. $(2^9 - 1)/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q4/5

406

QUESTION ID

S11-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $2^{10} - 1$

GRADING NOTE $(2^{10} - 1)/(2 - 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q4/5

Explain how the identity gives $a(1 - r^n)/(1 - r)$ for $a + ar + \ldots + ar^{n-1}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q5/5

407

QUESTION ID

S11-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Put x = r and multiply by a: $a(r^n - 1) = a(r - 1)(1 + r + \ldots + r^{n-1})$; divide by r − 1 (r ≠ 1) and rewrite signs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 2

Describe how partial sums behave when the ratio is between −1 and 1 • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

For the series 8 + 4 + 2 + 1 + ..., find S_5 and S_{10} to 4 decimal places.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q1/5

409

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$S_5 = 15.5000$;

$S_{10} = 15.9844$

GRADING NOTE Rounding to 4 decimal places. $S_n = 8(1 - 0.5^n)/0.5 = 16(1 - 0.5^n)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q1/5

Using $S_n = 16(1 - 0.5^n)$, explain why every partial sum of $8 + 4 + 2 + \ldots$ is less than 16 but can be made as close to 16 as you like.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q2/5

410

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.5^n is positive, so $S_n < 16$; 0.5^n shrinks toward 0 as n grows, so S_n gets as close to 16 as you like.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q2/5

True or false?

For r = 2, the partial sums of $a + ar + ar^2 + \ldots$ ($a > 0$) stay below a fixed number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q3/5

411

QUESTION ID

S11-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE r^n grows without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q3/5

For 10 − 5 + 2.5 − 1.25 + …, the partial sums get close to

A. 20/3

B. 15

C. 5

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q4/5

412

QUESTION ID

S11-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 20/3

GRADING NOTE a/(1 − r) = 10/1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q4/5

A ball travels 10 m down, then rebounds and falls in a geometric pattern so the total distance after n bounces is $10 + 2 \cdot 6(1 - 0.6^n)/0.4$. What total can the distance never reach, and why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q5/5

413

QUESTION ID

S11-F2-Q5

TYPE

Short answer

CORRECT ANSWER

40 m: $2 \cdot 6/0.4 = 30$ and $10 + 30 = 40$; $(1 - 0.6^n) < 1$ always.

GRADING NOTE Form required: the value and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q5/5

Use the finite geometric series formula to solve problems

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Identify the first term, ratio, and number of terms	**HSF-LE.A.2**
BACKWARD 2	Change a percent to a growth factor	**7.RP.A.3**
BACKWARD 3	Round results of calculations correctly	**HSN-Q.A.3**
FORWARD 1	Find the number of terms with logarithms	**HSF-LE.A.4**
FORWARD 2	Compare two payment plans with the series formula	**HSA-SSE.B.4**

SET 12 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 12 terms of 5 + 15 + 45 + ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q1/5

416

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

1,328,600

GRADING NOTE Any equivalent form. $5(3^{12} - 1)/(3 - 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q1/5

Leah deposits $250 at the end of each month into an account that earns 0.5% interest per month. The balance just after the 36th deposit is $250 + 250(1.005) + \ldots + 250(1.005)^{35}$. Find it to the nearest cent.

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q2/5

417

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

$9834.03

GRADING NOTE Rounding to the nearest cent required. $250(1.005^{36} - 1)/0.005 = 9834.03$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q2/5

A $180,000 loan is repaid with 360 equal monthly payments P at a monthly interest rate of 0.4%. P satisfies
$180{,}000 = P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$. Use the geometric series formula to find P to the nearest cent.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q3/5

QUESTION ID
S12-M-Q3

TYPE
Short answer

CORRECT ANSWER
$944.40

GRADING NOTE Rounding to the nearest cent required. a = P/1.004, r = 1/1.004, n = 360: P = 180,000 · $0.004/(1 - 1.004^{-360}) = 944.40$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q3/5

A ball is dropped from 10 m and each rebound reaches 60% of the previous height. Find the total vertical distance it has traveled when it hits the ground for the 6th time, to the nearest hundredth of a meter.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q4/5

419

QUESTION ID

S12-M-Q4

TYPE

Short answer

CORRECT ANSWER

37.67 m

GRADING NOTE Rounding to 0.01 required. $10 + 2(6 + 3.6 + 2.16 + 1.296 + 0.7776) = 10 + 2 \cdot 6(1 - 0.6^5)/0.4 = 37.6672$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q4/5

What is the least number of terms of 2 + 6 + 18 + ... whose sum is greater than 10,000? Show the sums that decide it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q5/5

420

QUESTION ID

S12-M-Q5

TYPE

Short answer

CORRECT ANSWER

9 terms: $S_8 = 6560$ and $S_9 = 19{,}682$

GRADING NOTE Form required: n and the two sums. $S_n = 3^n - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 1

Identify the first term, ratio, and number of terms • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

For 5 + 15 + 45 + ... (12 terms), give a, r, and n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q1/5

422

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

a = 5, r = 3, n = 12

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q1/5

For $250 + 250(1.005) + \ldots + 250(1.005)^{35}$, give a, r, and n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q2/5

423

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 250, r = 1.005, n = 36

GRADING NOTE Form required: all three. Exponents 0 to 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q2/5

For 6 + 3.6 + 2.16 + 1.296 + 0.7776, r is

A. 0.4

B. 1.67

C. 0.6

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q3/5

424

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.6

GRADING NOTE 3.6/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q3/5

True or false?

A list with exponents 0 through 35 has 35 terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q4/5

425

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 36 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q4/5

For $P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$, give the first term and the ratio.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q5/5

426

QUESTION ID

S12-B1-Q5

TYPE

Short answer

CORRECT ANSWER

a = P/1.004,

r = 1/1.004

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 2

Change a percent to a growth factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

What factor adds 0.5% interest?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q1/5

428

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.005

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A height is 60% of the one before. What is the ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q2/5

429

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

0.4% per month as a factor is

A. 1.04

B. 1.004

C. 0.004

D. 1.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q3/5

430

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.004

GRADING NOTE 1 + 0.004.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q3/5

True or false?

Interest of 12% per year is a factor of 0.12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q4/5

431

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The growth factor is 1 + 0.12 = 1.12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A loss of 15% per year is what factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q5/5

QUESTION ID

S12-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.85

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 3

Round results of calculations correctly • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Round 9834.0262 to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q1/5

434

QUESTION ID

S12-B3-Q1

TYPE

Short answer

CORRECT ANSWER

9834.03

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 37.6672 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q2/5

435

QUESTION ID

S12-B3-Q2

TYPE

Short answer

CORRECT ANSWER

37.67

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q2/5

True or false?

Rounding 1.005 to 1.01 before raising it to the 36th power gives the same answer to the nearest cent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q3/5

436

QUESTION ID

S12-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Early rounding changes the result a lot; round only at the end.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q3/5

1.005^{36} is closest to

A. 1.1967

B. 1.18

C. 1.2

D. 1.036

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q4/5

437

QUESTION ID

S12-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1.1967

GRADING NOTE Calculator value 1.19668....

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 3^{12} exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q5/5

438

QUESTION ID

S12-B3-Q5

TYPE

Short answer

CORRECT ANSWER

531441

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 1

Find the number of terms with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

How many end-of-month deposits of $250 at 0.5% per month are needed for the balance to reach $20,000? Use $250(1.005^n - 1)/0.005 \geq 20{,}000$.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q1/5

440

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

68 deposits

GRADING NOTE Form required: a whole number. $1.005^n \geq 1 + 20{,}000(0.005)/250 = 1.4$, so $n \geq \ln 1.4/\ln 1.005 \approx 67.46$; check: n = 67 gives about $19,838.70 and n = 68 about $20,187.89.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q1/5

Solve $2(3^n - 1)/2 = 3^n - 1 = 1{,}000{,}000$ for n exactly, then give the least whole number of terms whose sum is at least 1,000,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q2/5

441

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$n = \log_3(1{,}000{,}001)$

≈ 12.58, so 13 terms

GRADING NOTE Form required: the exact value and the whole number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q2/5

True or false?

For a ratio greater than 1, more terms always give a larger sum when a > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q3/5

442

QUESTION ID
S12-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each added term is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q3/5

The sum $5(2^n - 1)$ first exceeds 5,000 when n =

A. 9

B. 11

C. 10

D. 1000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q4/5

443

QUESTION ID

S12-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE $2^n > 1001$; $2^{10} = 1024$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q4/5

Explain why the number of terms must be rounded up, not to the nearest whole number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q5/5

444

QUESTION ID

S12-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The sum must reach the target; rounding down would give a sum that is still too small.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 2

Compare two payment plans with the series formula • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Plan A pays $1,000 per month for 24 months. Plan B pays $500 the first month and 8% more each month after that, for 24 months. Which plan pays more in total, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q1/5

446

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Plan B: $500(1.08^{24} - 1)/0.08 \approx \$33{,}382.38$ vs $24,000; about $9,382.38 more

GRADING NOTE Form required: the plan and the difference to the nearest cent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q1/5

Plan A pays $1,000 per month. Plan B pays $500 the first month and 8% more each month after that. After how many months does Plan B's total first exceed Plan A's total for the same number of months?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q2/5

447

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

After 18 months

GRADING NOTE Form required: the month. Compare $500(1.08^n - 1)/0.08$ with $1000n$: $n = 17$ gives $16{,}875.11 < 17{,}000$; $n = 18$ gives $18{,}725.12 > 18{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q2/5

True or false?

A geometric plan with ratio greater than 1 eventually pays more in total than any plan with a fixed monthly amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q3/5

448

QUESTION ID
S12-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its terms grow without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q3/5

Plan C pays $2,000 then 5% less each month. Its total over 24 months is closest to

A. $40,000

B. $24,000

C. $48,000

D. $28,320

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q4/5

449

QUESTION ID

S12-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $28,320

GRADING NOTE $2000(1 - 0.95^{24})/0.05 \approx 28{,}320$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q4/5

Name one assumption in comparing the plans by their totals only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q5/5

450

QUESTION ID
S12-F2-Q5

TYPE
Short answer

CORRECT ANSWER
For example: money received later is treated as equal in value to money now (no interest or inflation).

GRADING NOTE Any reasonable assumption.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q5/5

Add and subtract polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Combine like terms with the same power	**7.EE.A.1**
BACKWARD 2	Subtract a whole expression	**6.EE.A.3**
BACKWARD 3	Write polynomials in standard form	**HSA-SSE.A.1.a**
FORWARD 1	Reason about the degree of a sum or difference	**HSA-APR.A.1**
FORWARD 2	Add and subtract rational expressions	**HSA-APR.D.7**

SET 13 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Simplify $(4x^3 - 2x^2 + 7) + (-x^3 + 5x - 9)$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q1/5

453

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^3 - 2x^2 + 5x - 2$

GRADING NOTE Form required: standard form (descending powers). $4x^3 - x^3 = 3x^3$; $-2x^2$; $5x$; $7 - 9 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q1/5

Simplify $(6x^4 - x^3 + 2x) - (2x^4 + 3x^3 - x^2 + 2x - 1)$.
Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q2/5

454

QUESTION ID
S13-M-Q2

TYPE
Short answer

CORRECT ANSWER
$4x^4 - 4x^3 + x^2 + 1$

GRADING NOTE Form required: standard form. The x terms cancel: $2x - 2x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q2/5

A company's revenue is $R(x) = -0.01x^3 + 3x^2 + 40x$ dollars and its cost is $C(x) = 0.02x^2 + 15x + 900$ dollars for x items. Write the profit $P(x) = R(x) - C(x)$ in standard form.

$P(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q3/5

455

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

$-0.01x^3 + 2.98x^2 + 25x - 900$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q3/5

Simplify $(x^5 - 3x^3 + x) - (x^5 + 2x^4 - 3x^3 - 7)$. Write the result in standard form and give its degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q4/5

456

QUESTION ID

S13-M-Q4

TYPE

Short answer

CORRECT ANSWER

$-2x^4 + x + 7$; degree 4

GRADING NOTE Form required: standard form and the degree. The x^5 and x^3 terms cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q4/5

$p(x) = 5x^3 + x^2 - 6$. Find the polynomial $q(x)$ for which $p(x) + q(x) = 2x^3 - x + 4$.

$q(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q5/5

457

QUESTION ID

S13-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-3x^3 - x^2 - x + 10$

GRADING NOTE Form required: standard form. $q = (2x^3 - x + 4) - (5x^3 + x^2 - 6)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 1

Combine like terms with the same power • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $4x^3 - x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

459

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $-2x^2 + 5x - 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

460

QUESTION ID

S13-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-5x^2 + 5x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

True or false?

$x^3 + x^2 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

461

QUESTION ID

S13-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x^3 and x^2 are unlike terms; they cannot be combined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

Which pair are like terms?

A. $4x^2$ and $-x^2$

B. $4x^2$ and $4x$

C. x^3 and $3x$

D. x and 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

462

QUESTION ID

S13-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $4x^2$ and $-x^2$

GRADING NOTE Same variable, same power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $7 - 9 + 2x - 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

463

QUESTION ID

S13-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 2

Subtract a whole expression • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $-(2x^4 + 3x^3 - x^2 + 2x - 1)$ without parentheses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q1/5

465

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$-2x^4 - 3x^3 + x^2 - 2x + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q1/5

Simplify (5 − x) − (3 − 4x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q2/5

466

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2 + 3x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q2/5

True or false?

$a - (b - c) = a - b - c.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q3/5

467

QUESTION ID

S13-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $a - (b - c) = a - b + c.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q3/5

$-(x^2 - 7)$ equals

A. $x^2 + 7$

B. $-x^2 - 7$

C. $-x^2 + 7$

D. $x^2 - 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q4/5

468

QUESTION ID

S13-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-x^2 + 7$

GRADING NOTE Distribute -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify 10 − (x + 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q5/5

469

QUESTION ID

S13-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6 − x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 3

Write polynomials in standard form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Write $7 + 5x - 2x^3 + x^2$ in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q1/5

471

QUESTION ID

S13-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$-2x^3 + x^2 + 5x + 7$

GRADING NOTE Form required: descending powers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q1/5

What is the leading coefficient of $3 - 4x^2 + x^4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q2/5

472

QUESTION ID

S13-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q2/5

The degree of $-2x^4 + x + 7$ is

A. 7

B. 1

C. 4

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q3/5

473

QUESTION ID

S13-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE Highest power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q3/5

True or false?

The standard form of $2x - x^3$ is $2x - x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q4/5

474

QUESTION ID

S13-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Standard form lists powers in descending order: $-x^3 + 2x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q4/5

What is the constant term of $6x^3 - 2x + 9$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q5/5

475

QUESTION ID

S13-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 1

Reason about the degree of a sum or difference • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

p(x) has degree 4 and q(x) has degree 4. What degrees can p(x) + q(x) have? Give an example where the degree drops.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q1/5

477

QUESTION ID
S13-F1-Q1

TYPE
Short answer

CORRECT ANSWER
4 or less (or the zero polynomial); for example $(x^4 + x) + (-x^4 + 1) = x + 1$

GRADING NOTE Any correct example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q1**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q1/5

For which value of k does
$(kx^3 + 2x - 1) - (5x^3 - x^2 + 3)$ have degree 2?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q2/5

478

QUESTION ID
S13-F1-Q2

TYPE
Short answer

CORRECT ANSWER
5

GRADING NOTE Any equivalent form. The x^3 terms cancel when k = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q2/5

True or false?

The difference of two polynomials is always a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q3/5

479

QUESTION ID
S13-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Polynomials are closed under subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q3/5

p has degree 5 and q has degree 3. The degree of $p - q$ is

A. 5

B. 2

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q4/5

480

QUESTION ID
S13-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 5

GRADING NOTE The x^5 term of p cannot cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q4**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q4/5

Explain why the sum of two polynomials is a polynomial.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q5/5

481

QUESTION ID
S13-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Adding like terms gives terms that are a coefficient times a whole-number power of x, which is the form of a polynomial.

GRADING NOTE Any equivalent explanation (closure under addition).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 2

Add and subtract rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Write 3/x + 2/(x + 1) as one rational expression, and state the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q1/5

483

QUESTION ID
S13-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(5x + 3)/(x(x + 1)); x ≠ 0, −1

GRADING NOTE Form required: one fraction and the exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q1/5

Write 1/(x − 2) − 1/(x + 2) as one rational expression, and state the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q2/5

484

QUESTION ID
S13-F2-Q2

TYPE
Short answer

CORRECT ANSWER
4/((x − 2)(x + 2)); x ≠ 2, −2

GRADING NOTE Form required: one fraction and the exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q2**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q2/5

True or false?

Polynomials are closed under division.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q3/5

485

QUESTION ID
S13-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For example 1 ÷ x is not a polynomial; rational expressions are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

x/2 + x/3 equals

A. 5x/5

B. x/5

C. 5x/6

D. 2x/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q4/5

486

QUESTION ID
S13-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 5x/6

GRADING NOTE Common denominator 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q4/5

Explain how adding rational expressions is like adding fractions of integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q5/5

487

QUESTION ID
S13-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Rewrite both over a common denominator, then add the numerators; polynomials play the role of integers.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q5/5

Multiply polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Distribute a monomial over a polynomial	**HSA-APR.A.1**
BACKWARD 2	Multiply powers with the same base	**8.EE.A.1**
BACKWARD 3	Multiply two binomials	**HSA-APR.A.1**
FORWARD 1	Find a coefficient of a product without full expansion	**HSA-APR.A.1**
FORWARD 2	Expand powers of binomials with the Binomial Theorem	**HSA-APR.C.5**

SET 14 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(x^2 + 3x - 2)(2x - 5)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q1/5

490

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

$2x^3 + x^2 - 19x + 10$

GRADING NOTE Form required: standard form. $2x^3 - 5x^2 + 6x^2 - 15x - 4x + 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q1/5

Expand $(x - 2)^3$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q2/5

491

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 - 6x^2 + 12x - 8$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q2/5

Multiply $(x^2 - x + 1)(x^2 + x + 1)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q3/5

492

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x^4 + x^2 + 1$

GRADING NOTE Form required: standard form. $((x^2 + 1) - x)((x^2 + 1) + x) = (x^2 + 1)^2 - x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q3/5

Multiply (2x + 1)(x − 3)(x + 4). Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q4/5

493

QUESTION ID

S14-M-Q4

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 3x^2 - 23x - 12$

GRADING NOTE Form required: standard form. $(x - 3)(x + 4) = x^2 + x - 12$; times $(2x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q4/5

A box has length x + 2, width x − 1, and height 2x + 3 (in cm). Write its volume as a polynomial in standard form.

V(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q5/5

494

QUESTION ID

S14-M-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 5x^2 - x - 6$

GRADING NOTE Form required: standard form. $(x^2 + x - 2)(2x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 1

Distribute a monomial over a polynomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $2x(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q1/5

496

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 6x^2 - 4x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q1/5

Multiply $-5(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q2/5

497

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-5x^2 - 15x + 10$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q2/5

True or false?

$x^2(x + 4) = x^3 + 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q3/5

498

QUESTION ID

S14-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 \cdot 4 = 4x^2$, so the product is $x^3 + 4x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q3/5

$-3x(x - 1)$ equals

A. $3x^2 - 3x$

B. $-3x^2 + 3x$

C. $-3x^2 - 3x$

D. $-3x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q4/5

499

QUESTION ID

S14-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $-3x^2 + 3x$

GRADING NOTE $-3x \cdot -1 = 3x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $x(x^2 - x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q5/5

500

QUESTION ID

S14-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^3 - x^2 + x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 2

Multiply powers with the same base • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $x^2 \cdot x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q1/5

502

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x^5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $2x \cdot 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q2/5

503

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$6x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q2/5

True or false?

$x^2 \cdot x^2 = x^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q3/5

504

QUESTION ID

S14-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$-x \cdot x^3$ equals

A. x^4

B. $-x^3$

C. $-x^4$

D. $-x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q4/5

505

QUESTION ID

S14-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-x^4$

GRADING NOTE $-1 \cdot x^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(x^2)(-4x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q5/5

506

QUESTION ID

S14-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$-4x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 3

Multiply two binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x - 3)(x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q1/5

508

QUESTION ID

S14-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + x - 12$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x + 2)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q2/5

509

QUESTION ID

S14-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + x - 2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - 2)^2$ equals

A. $x^2 + 4$

B. $x^2 - 4$

C. $x^2 - 4x + 4$

D. $x^2 - 2x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q3/5

510

QUESTION ID

S14-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2 - 4x + 4$

GRADING NOTE Square of a binomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q3/5

True or false?

$(2x + 1)(x - 3) = 2x^2 - 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q4/5

511

QUESTION ID

S14-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The middle terms $-6x + x = -5x$ are missing: $2x^2 - 5x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q4/5

Multiply $(2x + 3)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q5/5

512

QUESTION ID

S14-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2 + x - 3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 1

Find a coefficient of a product without full expansion • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Find the coefficient of x^2 in $(x^2 + 3x - 2)(4x^2 - x + 5)$ without expanding everything.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q1/5

514

QUESTION ID
S14-F1-Q1

TYPE
Short answer

CORRECT ANSWER
−6

GRADING NOTE Any equivalent form. x^2 terms: $x^2 \cdot 5 + 3x \cdot (-x) + (-2) \cdot 4x^2 = 5 - 3 - 8 = -6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q1/5

For which k does $(x + k)(x^2 - 3x + 1)$ have no x^2 term?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q2/5

515

QUESTION ID
S14-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. x^2 coefficient: $-3 + k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q2/5

True or false?

The constant term of a product of polynomials is the product of their constant terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q3/5

516

QUESTION ID
S14-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Only constants multiply to a constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q3/5

The leading coefficient of $(2x^3 - 1)(5x^2 + x)$ is

A. 7

B. 5

C. 2

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q4/5

517

QUESTION ID
S14-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 10

GRADING NOTE $2 \cdot 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q4**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q4/5

Find the coefficient of x in (x − 1)(x − 2)(x − 3).

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q5/5

518

QUESTION ID
S14-F1-Q5

TYPE
Short answer

CORRECT ANSWER
11

GRADING NOTE Any equivalent form. 1·2 + 1·3 + 2·3 = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q5**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 2

Expand powers of binomials with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 1)^4$ using Pascal's Triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q1/5

520

QUESTION ID
S14-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^4 + 4x^3 + 6x^2 + 4x + 1$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q1/5

Expand $(x - 2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q2/5

521

QUESTION ID
S14-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$x^4 - 8x^3 + 24x^2 - 32x + 16$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

The coefficients of $(x + y)^5$ are 1, 5, 10, 5, 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q3/5

522

QUESTION ID
S14-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Row 5 of Pascal's Triangle has six entries: 1, 5, 10, 10, 5, 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q3/5

The coefficient of x^2y^2 in $(x + y)^4$ is

A. 2

B. 6

C. 4

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q4/5

523

QUESTION ID
S14-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 6

GRADING NOTE C(4, 2) = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q4/5

Explain why the coefficients of $(x + y)^n$ are symmetric.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q5/5

524

QUESTION ID
S14-F2-Q5

TYPE
Short answer

CORRECT ANSWER
C(n, k) = C(n, n − k): choosing k of the n factors to give x is the same as choosing the n − k that give y.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q5/5

Find remainders with the Remainder Theorem

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Evaluate a polynomial function	**HSF-IF.A.2**
BACKWARD 2	Powers of negative numbers	**7.NS.A.2**
BACKWARD 3	Dividend = divisor × quotient + remainder	**4.NBT.B.6**
FORWARD 1	Find unknown coefficients from given remainders	**HSA-APR.B.2**
FORWARD 2	Prove the Remainder Theorem	**HSA-APR.B.2**

SET 15 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Use the Remainder Theorem to find the remainder when $p(x) = x^3 - 4x^2 + 2x - 5$ is divided by $x - 3$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q1/5

527

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form. $p(3) = 27 - 36 + 6 - 5 = -8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q1/5

Use the Remainder Theorem to find the remainder when $p(x) = 2x^4 + x^3 - 3x + 7$ is divided by $x + 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q2/5

528

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

37

GRADING NOTE Any equivalent form. Divisor $x - (-2)$: $p(-2) = 32 - 8 + 6 + 7 = 37$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q2/5

Use the Remainder Theorem to find the remainder when $p(x) = x^{100} + 3$ is divided by $x + 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q3/5

529

QUESTION ID

S15-M-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $p(-1) = (-1)^{100} + 3 = 1 + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-M-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q3/5

Use the Remainder Theorem to find the remainder when $p(x) = 4x^3 - 6x^2 + x - 1$ is divided by $x - 1/2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q4/5

530

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form (−1.5). p(1/2) = 1/2 − 3/2 + 1/2 − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q4/5

Use the Remainder Theorem to find the remainder when $p(x) = x^4 - 5x^2 + 4x$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q5/5

531

QUESTION ID

S15-M-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $p(2) = 16 - 20 + 8 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 1

Evaluate a polynomial function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = x^3 - 4x^2 + 2x - 5$. Find $p(1)$.

$p(1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q1/5

533

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q1/5

$p(x) = 2x^2 - x + 3$. Find $p(-2)$.

$p(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q2/5

534

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. 8 + 2 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q2/5

True or false?

If $p(x) = x^2 + 1$, then $p(-3) = -8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q3/5

535

QUESTION ID

S15-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-3)^2 + 1 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q3/5

If $p(x) = x^3 - x$, then $p(2) =$

A. 4

B. 6

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q4/5

536

QUESTION ID

S15-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE 8 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q4/5

$p(x) = 4x - 1$. Find $p(1/2)$.

$p(1/2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q5/5

537

QUESTION ID

S15-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 2

Powers of negative numbers • Grade 7 • The Number System • 7.NS.A.2

Evaluate $(-2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q1/5

539

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q1/5

Evaluate $(-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q2/5

540

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q2/5

$(-1)^{100}$ equals

A. -100

B. -1

C. 100

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q3/5

541

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE An even power of -1 is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q3/5

True or false?

$-2^4 = (-2)^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q4/5

542

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-2^4 = -16$ but $(-2)^4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $2(-2)^4 + (-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q5/5

543

QUESTION ID

S15-B2-Q5

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. 32 − 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 3

Dividend = divisor × quotient + remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Write 47 = 5 × q + r with 0 ≤ r < 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

545

QUESTION ID

S15-B3-Q1

TYPE

Short answer

CORRECT ANSWER

47 = 5 × 9 + 2

GRADING NOTE Form required: q = 9 and r = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

A division gives quotient 12 and remainder 3 with divisor 7. What was the dividend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

546

QUESTION ID

S15-B3-Q2

TYPE

Short answer

CORRECT ANSWER

87

GRADING NOTE Any equivalent form. 7 × 12 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

True or false?

If a division leaves remainder 0, the divisor is a factor of the dividend.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

547

QUESTION ID

S15-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Dividend = divisor × quotient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

$x^3 - 1 = (x - 1)(x^2 + x + 1) + 0$. The remainder on division by $x - 1$ is

A. 0

B. 1

C. $x^2 + x + 1$

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

548

QUESTION ID

S15-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0

GRADING NOTE Read from the identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

$p(x) = (x - 3)(x^2 - x - 1) - 8$. What is the remainder when $p(x)$ is divided by $x - 3$?

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

549

QUESTION ID

S15-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 1

Find unknown coefficients from given remainders • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

When $p(x) = x^3 + kx^2 - 4$ is divided by $x - 2$, the remainder is 12. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q1/5

551

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. $p(2) = 8 + 4k - 4 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = x^3 + ax + b$ leaves remainder 5 on division by $x - 1$ and remainder -1 on division by $x + 1$. Find a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q2/5

552

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 2, b = 2

GRADING NOTE Form required: both. $1 + a + b = 5$ and $-1 - a + b = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q2/5

True or false?

If p(x) leaves remainder 0 on division by x − 4, then p(−4) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q3/5

553

QUESTION ID

S15-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The remainder on division by x − 4 is p(4), so p(4) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q3/5

$p(x) = 2x^2 + kx + 1$ leaves remainder 7 on division by $x + 1$. Then $k =$

A. 4

B. -4

C. -6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q4/5

554

QUESTION ID

S15-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −4

GRADING NOTE $2 - k + 1 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q4/5

For which k is the remainder of $x^4 - kx + 6$ divided by $x + 2$ equal to 0?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q5/5

555

QUESTION ID

S15-F1-Q5

TYPE

Short answer

CORRECT ANSWER

−11

GRADING NOTE Any equivalent form. 16 + 2k + 6 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 2

Prove the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Which argument proves that the remainder when a polynomial p(x) is divided by x − a is p(a)?

A. Write p(x) = (x − a)q(x) + r, where r is a constant because it has degree less than 1. Substituting x = a gives p(a) = 0 · q(a) + r = r.

B. Dividing x^3 by x − 2 leaves remainder 8, and $2^3 = 8$, so the remainder is always p(a).

C. Since x − a = 0 exactly when x = a, dividing p(x) by x − a is the same as replacing x with a, so the remainder must be p(a).

D. Write p(x) = (x − a)q(x) + r. Substituting x = 0 gives p(0) = (0 − a)q(0) + r = −a · q(0) + r, so the remainder r equals p(a) for every a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q1/5

QUESTION ID

S15-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Write p(x) = (x − a)q(x) + r, where r is a constant because it has degree less than 1. Substituting x = a gives p(a) = 0 · q(a) + r = r.

GRADING NOTE The division identity with a constant remainder, then x = a makes the factor (a − a) zero. The others check one example, substitute the wrong value (x = 0 gives p(0), not p(a)), or state the result without a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q1/5

Why must the remainder on division by x − a be a constant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q2/5

558

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The remainder has a lower degree than the divisor x − a, which has degree 1, so it has degree 0 (a constant).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q2/5

True or false?

The Remainder Theorem also works for division by 2x − 1 with remainder p(1/2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q3/5

559

QUESTION ID

S15-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE p(x) = (2x − 1)q(x) + r gives p(1/2) = r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q3/5

Which step uses the fact that a − a = 0?

A. Graphing p(x)

B. Dividing p(x) by x

C. Factoring p(x)

D. Substituting x = a into p(x) = (x − a)q(x) + r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q4/5

560

QUESTION ID

S15-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Substituting x = a into p(x) = (x − a)q(x) + r

GRADING NOTE The product term vanishes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q4/5

Explain how the proof shows that p(a) = 0 exactly when x − a is a factor of p(x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q5/5

561

QUESTION ID

S15-F2-Q5

TYPE

Short answer

CORRECT ANSWER

p(a) = r, and r = 0 exactly when p(x) = (x − a)q(x), i.e. x − a is a factor.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q5/5

Use p(a) = 0 to find factors

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Use the Remainder Theorem	**HSA-APR.B.2**
BACKWARD 2	Factor the quadratic quotient	**HSA-SSE.A.2**
BACKWARD 3	Divide a polynomial by a linear factor	**HSA-APR.A.1**
FORWARD 1	Choose parameters so that a given binomial is a factor	**HSA-APR.B.2**
FORWARD 2	Finish factorizations over the complex numbers	**HSN-CN.C.8**

SET 16 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Show that $x - 2$ is a factor of $p(x) = x^3 - 3x^2 - 4x + 12$, then factor $p(x)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q1/5

564

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

$p(2) = 0$; $p(x) = (x - 2)(x + 2)(x - 3)$

GRADING NOTE Form required: the check and the complete factorization. $p(2) = 8 - 12 - 8 + 12 = 0$; quotient $x^2 - x - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q1/5

Show that x + 1 is a factor of $p(x) = x^3 + 2x^2 - 5x - 6$, then factor p(x) completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q2/5

565

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

p(−1) = 0; p(x) = (x + 1)(x + 3)(x − 2)

GRADING NOTE Form required: the check and the factorization. Quotient $x^2 + x - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-M-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q2/5

Show that $x + 3$ is a factor of $p(x) = 2x^3 + 5x^2 - 4x - 3$, then factor $p(x)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q3/5

566

QUESTION ID

S16-M-Q3

TYPE

Short answer

CORRECT ANSWER

$p(-3) = 0$; $p(x) = (x + 3)(2x + 1)(x - 1)$

GRADING NOTE Form required: the check and the factorization. $p(-3) = -54 + 45 + 12 - 3 = 0$; quotient $2x^2 - x - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q3/5

Is $x - 4$ a factor of $p(x) = x^3 - 5x^2 + 2x + 9$? Justify with the Remainder Theorem.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q4/5

567

QUESTION ID

S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

No: $p(4) = 1 \neq 0$

GRADING NOTE Form required: the decision and $p(4)$. $64 - 80 + 8 + 9 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-M-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q4/5

$p(x) = x^3 - 7x + 6$. Use $p(a) = 0$ to find all the zeros of p, and write $p(x)$ in factored form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q5/5

568

QUESTION ID

S16-M-Q5

TYPE

Short answer

CORRECT ANSWER

Zeros 1, 2, −3; $p(x) = (x - 1)(x - 2)(x + 3)$

GRADING NOTE Form required: all zeros and the factorization. $p(1) = 0$, then the quotient $x^2 + x - 6 = (x - 2)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 1

Use the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder when $x^3 - 3x^2 - 4x + 12$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q1/5

570

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q1/5

Find the remainder when $x^3 + 2x^2 - 5x - 6$ is divided by $x - 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q2/5

571

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q2/5

True or false?

Dividing by x + 3 means evaluating at x = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q3/5

572

QUESTION ID

S16-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x + 3 = x − (−3): evaluate at −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B1-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q3/5

The remainder of p(x) ÷ (x + 1) is

A. −p(1)

B. p(−1)

C. p(0)

D. p(1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q4/5

573

QUESTION ID

S16-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. p(−1)

GRADING NOTE x − (−1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $p(-3)$ for $p(x) = 2x^3 + 5x^2 - 4x - 3$.

$p(-3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q5/5

574

QUESTION ID
S16-B1-Q5

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 2

Factor the quadratic quotient • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q1/5

576

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 2)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $2x^2 - x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q2/5

577

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(2x + 1)(x − 1)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q2/5

$x^2 + x - 6$ equals

A. $(x - 3)(x + 2)$

B. $(x - 6)(x + 1)$

C. $(x + 3)(x - 2)$

D. $(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q3/5

578

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (x + 3)(x − 2)

GRADING NOTE Product −6, sum 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x^2 + 4$ factors over the integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q4/5

579

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q5/5

580

QUESTION ID

S16-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - 3)(x + 3)$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 3

Divide a polynomial by a linear factor • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find q(x) if $x^3 - 3x^2 - 4x + 12 = (x - 2)\,q(x)$.

q(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q1/5

582

QUESTION ID

S16-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 - x - 6$

GRADING NOTE Any equivalent form. Check by multiplying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q1/5

Multiply $(x + 1)(x^2 + x - 6)$ to check a division.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q2/5

583

QUESTION ID

S16-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + 2x^2 - 5x - 6$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q2/5

True or false?

If $(x - 1)(x^2 + x - 6) = x^3 - 7x + 6$, then $x^2 + x - 6$ is the quotient of $x^3 - 7x + 6 \div (x - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q3/5

584

QUESTION ID

S16-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of the quotient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^3 - 8 \div (x - 2)$ equals

A. $x^2 + 2x + 4$

B. $x^2 + 4$

C. $x^2 - 4$

D. $x^2 - 2x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q4/5

585

QUESTION ID

S16-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 2x + 4$

GRADING NOTE Difference of cubes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find q(x) if $2x^3 + 5x^2 - 4x - 3 = (x + 3)\,q(x)$.

q(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q5/5

586

QUESTION ID

S16-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2 - x - 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 1

Choose parameters so that a given binomial is a factor • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find k so that $x - 3$ is a factor of $x^3 + kx^2 - 3x + 9$.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q1/5

588

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form. $27 + 9k - 9 + 9 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q1/5

Find a and b so that $x - 1$ and $x + 2$ are both factors of $x^3 + ax^2 + bx - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q2/5

589

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 2, b = −1

GRADING NOTE Form required: both. $1 + a + b - 2 = 0$ and $-8 + 4a - 2b - 2 = 0$ give $a + b = 1$ and $2a - b = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q2/5

True or false?

If $x - 2$ and $x + 2$ are factors of $p(x)$, then $x^2 + 4$ is a factor of $p(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q3/5

590

QUESTION ID

S16-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x - 2)(x + 2) = x^2 - 4$, not $x^2 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q3/5

For which k is x + 1 a factor of $x^4 + kx + 3$?

A. 2

B. −4

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q4/5

591

QUESTION ID

S16-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE $1 - k + 3 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q4/5

Write a cubic with leading coefficient 1 that has $x - 1$, $x + 1$, and $x - 4$ as factors, in standard form.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q5/5

592

QUESTION ID

S16-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^3 - 4x^2 - x + 4$

GRADING NOTE Form required: standard form. $(x^2 - 1)(x - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 2

Finish factorizations over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

$x - 1$ is a factor of $x^3 - 1$. Factor $x^3 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q1/5

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1/2 - (\sqrt{3}/2)i)(x + 1/2 + (\sqrt{3}/2)i)$

GRADING NOTE Any equivalent form. The quotient $x^2 + x + 1$ has zeros $-1/2 \pm (\sqrt{3}/2)i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$x + 2$ is a factor of $x^3 + 2x^2 + 9x + 18$. Factor it completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q2/5

595

QUESTION ID
S16-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$(x + 2)(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q2/5

True or false?

Every factor found with the factor theorem gives a zero of the polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q3/5

596

QUESTION ID

S16-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x - a$ is a factor exactly when $p(a) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q3/5

$x - 2$ is a factor of $x^3 - 2x^2 + x - 2$. The other zeros are

A. i and −i

B. −2

C. 1 and −1

D. 2i and −2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q4/5

597

QUESTION ID

S16-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. i and −i

GRADING NOTE Quotient $x^2 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q4/5

$x - 3$ is a factor of $x^3 - 3x^2 + 4x - 12$. List all complex zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q5/5

598

QUESTION ID
S16-F2-Q5

TYPE
Short answer

CORRECT ANSWER
3, 2i, −2i

GRADING NOTE Form required: all three. Quotient $x^2 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q5/5

Identify zeros and multiplicities from a factorization

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Solve by the zero-product property	**HSA-REI.B.4.b**
BACKWARD 2	Factor to find hidden repeated factors	**HSA-SSE.A.2**
BACKWARD 3	An exponent counts repeated factors	**6.EE.A.1**
FORWARD 1	Build a polynomial from zeros and one more condition	**HSA-APR.B.3**
FORWARD 2	Decide what the zeros determine and what remains free	**HSA-APR.B.3**

SET 17 • HSA-APR.B.3

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $p(x) = (x - 4)^2(x + 1)(2x - 3)$ and give the multiplicity of each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q1/5

601

QUESTION ID
S17-M-Q1

TYPE
Short answer

CORRECT ANSWER
4 (multiplicity 2), −1 (multiplicity 1), 3/2 (multiplicity 1)

GRADING NOTE Form required: every zero with its multiplicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q1/5

Factor $p(x) = x^3 - 6x^2 + 9x$, then list its zeros with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q2/5

602

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

$p(x) = x(x - 3)^2$; 0 (multiplicity 1), 3 (multiplicity 2)

GRADING NOTE Form required: factorization and zeros with multiplicities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q2/5

List the zeros of $p(x) = (x^2 - 9)(x^2 + 2x + 1)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q3/5

603

QUESTION ID

S17-M-Q3

TYPE

Short answer

CORRECT ANSWER

3 (1), −3 (1), −1 (2)

GRADING NOTE Form required: every zero with its multiplicity. $(x - 3)(x + 3)(x + 1)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-M-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q3/5

List the zeros of $p(x) = -2x(x + 5)^3$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q4/5

604

QUESTION ID

S17-M-Q4

TYPE

Short answer

CORRECT ANSWER

0 (1), −5 (3)

GRADING NOTE Form required: every zero with its multiplicity. The constant −2 is not a zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q4/5

Factor $p(x) = x^4 - 8x^2 + 16$, then list its zeros with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q5/5

605

QUESTION ID

S17-M-Q5

TYPE

Short answer

CORRECT ANSWER

$p(x) = (x - 2)^2(x + 2)^2$; 2 (2), −2 (2)

GRADING NOTE Form required: factorization and zeros with multiplicities. $(x^2 - 4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 1

Solve by the zero-product property • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 4)(x + 1) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q1/5

607

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 or −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve x(2x − 3) = 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q2/5

608

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0 or 3/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q2/5

True or false?

If $(x - 2)^2 = 0$, then $x = 2$ or $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q3/5

609

QUESTION ID

S17-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solutions of $(x + 5)(x - 5) = 0$ are

A. 25

B. −5 and 5

C. 0 and 5

D. 5 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q4/5

610

QUESTION ID

S17-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −5 and 5

GRADING NOTE Each factor equal to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-2x(x + 5) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q5/5

611

QUESTION ID

S17-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0 or −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 2

Factor to find hidden repeated factors • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 6x + 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q1/5

613

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q1/5

Factor $x^2 + 2x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q2/5

614

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^4 - 8x^2 + 16$ equals

A. $(x^2 + 4)^2$

B. $(x^2 - 4)^2$

C. $(x^2 - 8)^2$

D. $(x - 4)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q3/5

615

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x^2 - 4)^2$

GRADING NOTE $u^2 - 8u + 16$ with $u = x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x^3 - 6x^2 + 9x = x(x + 3)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q4/5

616

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x(x + 3)^2 = x^3 + 6x^2 + 9x$; the correct form is $x(x - 3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q5/5

617

QUESTION ID

S17-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q5**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 3

An exponent counts repeated factors • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $(x - 4)^2$ as a product without an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q1/5

619

QUESTION ID

S17-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 4)(x − 4)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many factors of $(x + 5)$ are in $(x + 5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q2/5

620

QUESTION ID

S17-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q2/5

True or false?

$(x - 2)^2(x - 2) = (x - 2)^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q3/5

621

QUESTION ID

S17-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q3/5

In $7(x - 1)^4$, the factor $x - 1$ appears

A. 28 times

B. 4 times

C. 1 time

D. 7 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q4/5

622

QUESTION ID

S17-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4 times

GRADING NOTE Exponent 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (x + 1)(x + 1)(x − 3) with an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q5/5

623

QUESTION ID

S17-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2(x - 3)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 1

Build a polynomial from zeros and one more condition • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Write the cubic polynomial with zeros −1, 2, and 4 (each multiplicity 1) whose graph passes through (0, 16).

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q1/5

625

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2(x + 1)(x − 2)(x − 4)

GRADING NOTE Any equivalent form. a(1)(−2)(−4) = 8a = 16 gives a = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q1/5

Write a polynomial of degree 3 with a zero of multiplicity 2 at x = 3, a zero at x = −2, and leading coefficient −1.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q2/5

626

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$-(x-3)^2(x+2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q2/5

True or false?

Zeros alone determine a polynomial completely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q3/5

627

QUESTION ID

S17-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any nonzero constant multiple has the same zeros; a further condition fixes the scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q3/5

A quartic has zeros 1 and −1, each of multiplicity 2, and p(0) = 3. Then p(x) =

A. $-3(x-1)^2(x+1)^2$

B. $3(x^2-1)$

C. $(x-1)^2(x+1)^2$

D. $3(x-1)^2(x+1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q4/5

628

QUESTION ID

S17-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $3(x-1)^2(x+1)^2$

GRADING NOTE $a \cdot 1 \cdot 1 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q4/5

A polynomial has zeros 0 (multiplicity 2) and 5 (multiplicity 1), degree 3, and p(1) = −8. Find p(x).

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q5/5

629

QUESTION ID

S17-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2(x - 5)$

GRADING NOTE Any equivalent form. a · 1 · (−4) = −8 gives a = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 2

Decide what the zeros determine and what remains free • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

All polynomials with zeros 1 and 3 only (each multiplicity 1) and degree 2 have what form?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q1/5

631

QUESTION ID
S17-F2-Q1

TYPE
Short answer

CORRECT ANSWER
a(x − 1)(x − 3) with any real a ≠ 0

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q1/5

If the degree is not fixed, give two different polynomials whose only real zeros are 1 and 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q2/5

632

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

For example $(x - 1)(x - 3)$ and $(x - 1)^2(x - 3)(x^2 + 1)$

GRADING NOTE Any two valid examples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q2/5

True or false?

Two cubic polynomials with the same three zeros (each multiplicity 1) must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q3/5

633

QUESTION ID

S17-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They can differ by a constant factor, for example $(x-1)(x-2)(x-3)$ and $2(x-1)(x-2)(x-3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q3/5

Which extra information fixes a cubic with zeros −2, 0, 2?

A. That it has degree 3

B. That 0 is a zero

C. Its end behavior only

D. Its value at x = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q4/5

634

QUESTION ID

S17-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Its value at x = 1

GRADING NOTE One point fixes the scale a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q4/5

Explain why knowing only the real zeros of a degree-4 polynomial may not determine it even up to scale.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q5/5

635

QUESTION ID

S17-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It may have non-real zeros (an irreducible quadratic factor such as $x^2 + c$) that the real zeros do not show, so different quartics share the same real zeros.

GRADING NOTE Any equivalent explanation with an example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q5/5

Construct a rough graph of a polynomial from its zeros

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Read zeros and multiplicities	**HSA-APR.B.3**
BACKWARD 2	End behavior from the leading term	**HSF-IF.B.4**
BACKWARD 3	Find the y-intercept by evaluating at 0	**HSF-IF.A.2**
BACKWARD 4	Sign of a power of a negative or positive number	**7.NS.A.2**
FORWARD 1	Sketch graphs from standard form by factoring first	**HSF-IF.C.7.c**
FORWARD 2	Write a polynomial from its rough graph	**HSA-APR.B.3**

SET 18 • HSA-APR.B.3

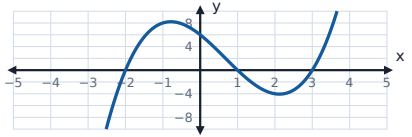
MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

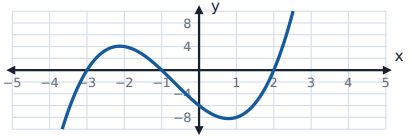
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = (x + 2)(x - 1)(x - 3)$?

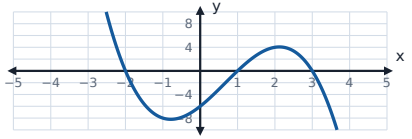
A



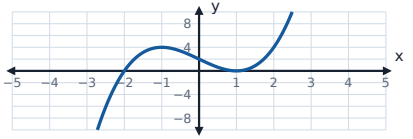
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q1/5

638

QUESTION ID

S18-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Zeros −2, 1, 3 (each crossed, multiplicity 1); y-intercept (2)(−1)(−3) = 6; degree 3 with positive leading coefficient, so it falls left and rises right. The others have the zeros with the wrong signs, the opposite end behavior, or a double zero at 1.

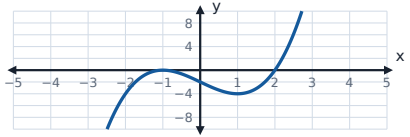
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q1/5

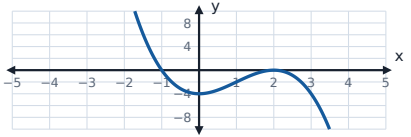
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = -(x + 1)^2(x - 2)$?

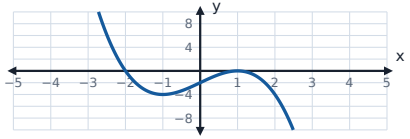
A



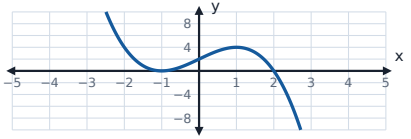
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q2/5

639

QUESTION ID

S18-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Double zero at −1 (touches), single zero at 2 (crosses); y-intercept $-(1)^2(-2) = 2$; leading coefficient −1 with odd degree, so it rises left and falls right. The others flip the sign, swap the multiplicities, or use the zeros 1 and −2.

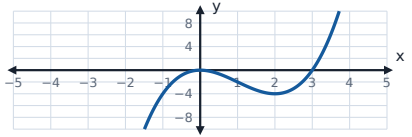
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q2/5

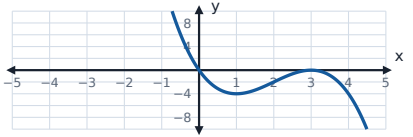
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = x(x - 3)^2$?

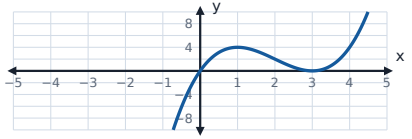
A



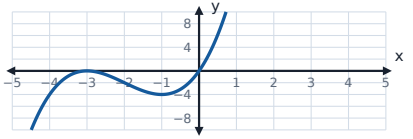
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q3/5

640

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Single zero at 0 (crosses), double zero at 3 (touches); y-intercept 0; $y(1) = 4 > 0$; falls left and rises right. The others swap the multiplicities, flip the sign, or put the double zero at -3.

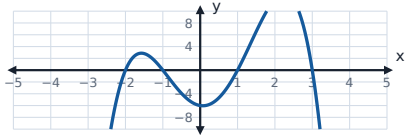
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q3/5

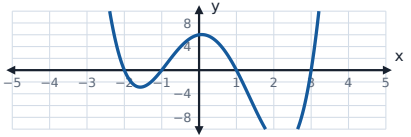
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = (x + 2)(x + 1)(x - 1)(x - 3)$?

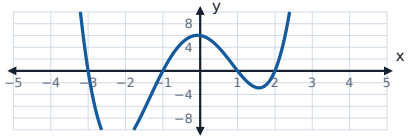
A



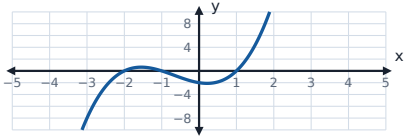
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q4/5

641

QUESTION ID

S18-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

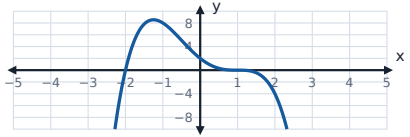
GRADING NOTE Four single zeros −2, −1, 1, 3; y-intercept (2)(1)(−1)(−3) = 6; degree 4 with positive leading coefficient, so it rises at both ends. The others are reflected, have zeros with the wrong signs (same y-intercept), or leave out a factor.

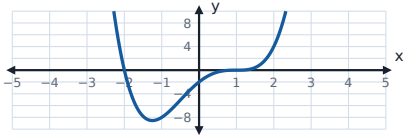
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

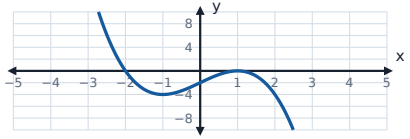
High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q4/5

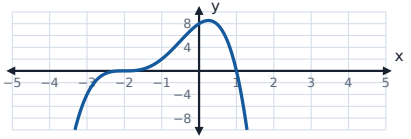
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = -(x - 1)^3(x + 2)$?

A 

B 

C 

D 

A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q5/5

642

QUESTION ID

S18-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Zero 1 has multiplicity 3 (crosses with a flattened shape), zero −2 is single; y-intercept $-(-1)^3(2) = 2$; degree 4 with negative leading coefficient, so it falls at both ends. The others flip the sign, use multiplicity 2 at 1, or put the triple zero at −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 1

Read zeros and multiplicities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q1/5

644

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$-2, 1, 3$

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q1/5

List the zeros of $-(x + 1)^2(x - 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q2/5

645

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−1 (2), 2 (1)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q2/5

True or false?

$x(x - 3)^2$ has a zero of multiplicity 2 at $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q3/5

646

QUESTION ID

S18-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 0$ has multiplicity 1; $x = 3$ has multiplicity 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q3/5

The zero $x = 1$ of $-(x - 1)^3(x + 2)$ has multiplicity

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q4/5

647

QUESTION ID

S18-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE Exponent 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the zeros of $(x + 2)(x + 1)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q5/5

QUESTION ID

S18-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2, −1, 1, 3

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 2

End behavior from the leading term • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

As x grows large and positive, what happens to x^3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q1/5

650

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

It grows large and positive.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

As x grows large and negative, what happens to $-x^4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q2/5

651

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

It becomes large and negative.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q2/5

The leading term of $(x + 2)(x - 1)(x - 3)$ is

A. $-x^3$

B. -6

C. $3x$

D. x^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q3/5

652

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x^3

GRADING NOTE $x \cdot x \cdot x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q3/5

True or false?

A polynomial of even degree with a positive leading coefficient falls at both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q4/5

653

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Like x^2, it rises at both ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the leading term of $-(x-1)^3(x+2)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q5/5

654

QUESTION ID

S18-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$-x^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 3

Find the y-intercept by evaluating at 0 • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find y when x = 0 for y = (x + 2)(x − 1)(x − 3).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q1/5

656

QUESTION ID

S18-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 2 · (−1) · (−3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q1/5

Find y when x = 0 for $y = -(x + 1)^2(x - 2)$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q2/5

657

QUESTION ID

S18-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. −(1)(−2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q2/5

True or false?

The y-intercept of $x(x - 3)^2$ is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q3/5

658

QUESTION ID

S18-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 0 (a factor x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q3/5

The y-intercept of $-(x-1)^3(x+2)$ is

A. 0

B. -2

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q4/5

659

QUESTION ID

S18-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $-(-1)(2) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q4/5

Find y when x = 1 for $y = x(x - 3)^2$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q5/5

660

QUESTION ID

S18-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 4

Sign of a power of a negative or positive number • Grade 7 • The Number System • 7.NS.A.2

Is $(-0.5)^2$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q1/5

662

QUESTION ID

S18-B4-Q1

TYPE

Short answer

CORRECT ANSWER

Positive

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q1/5

Is $(-0.5)^3$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q2/5

663

QUESTION ID

S18-B4-Q2

TYPE

Short answer

CORRECT ANSWER

Negative

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q2/5

True or false?

An even power of a negative number is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q3/5

664

QUESTION ID

S18-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pairs of negative factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q3/5

Near $x = 3$, $(x - 3)^2$ is

A. negative on both sides

B. positive on both sides

C. zero everywhere

D. positive on the left and negative on the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q4/5

665

QUESTION ID

S18-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. positive on both sides

GRADING NOTE A square does not change sign at $x = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q4/5

Does $(x - 1)^3$ change sign at $x = 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q5/5

666

QUESTION ID

S18-B4-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: negative for $x < 1$ and positive for $x > 1$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q5/5

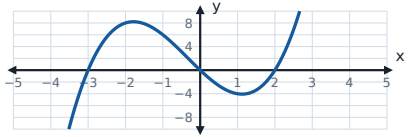
SET 18 • HSA-APR.B.3

FORWARD 1

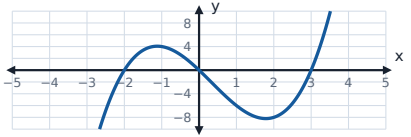
Sketch graphs from standard form by factoring first • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Factor $y = x^3 - x^2 - 6x$. Which graph is a rough graph of it?

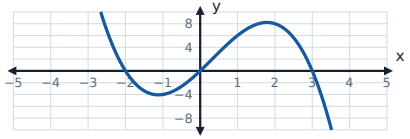
A



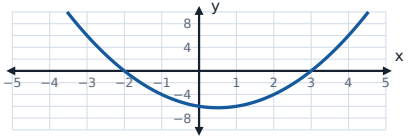
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q1/5

668

QUESTION ID

S18-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE $x^3 - x^2 - 6x = x(x - 3)(x + 2)$: crosses at -2, 0, 3; y-intercept 0; falls left and rises right. The others use $x(x + 3)(x - 2)$, the reflection, or drop the factor x.

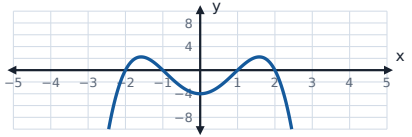
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q1/5

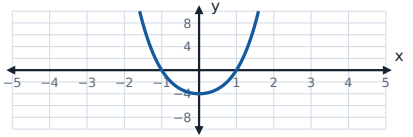
ANSWER KEY Reduced copy of the question page, with its answer

Factor $y = x^4 - 5x^2 + 4$. Which graph is a rough graph of it?

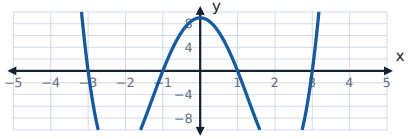
A



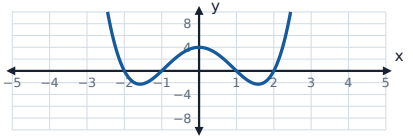
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q2/5

669

QUESTION ID

S18-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2)$: crosses at ±1 and ±2; y-intercept 4; rises at both ends. The others are reflected, factor as $(x^2 - 1)(x^2 + 4)$, or use $(x^2 - 1)(x^2 - 9)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q2/5

True or false?

$y = x^3 + 4x$ crosses the x-axis at $x = -2$, 0, and 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q3/5

670

QUESTION ID

S18-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x(x^2 + 4) = 0$ only at $x = 0$; $x^2 + 4$ has no real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q3/5

$y = -x^3 + 9x$ crosses the x-axis at

A. 0 and 9

B. 3 only

C. −3, 0, 3

D. −9, 0, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q4/5

671

QUESTION ID

S18-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −3, 0, 3

GRADING NOTE $-x(x - 3)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q4/5

$y = x^3 - 3x^2 + 4 = (x + 1)(x - 2)^2$. Describe its behavior at each zero.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q5/5

672

QUESTION ID

S18-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Crosses at x = −1; touches (turns) at x = 2.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

FORWARD 2

Write a polynomial from its rough graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A cubic's graph crosses the x-axis at −3 and touches it at 2, and passes through (0, 12). Write the polynomial.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q1/5

674

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(x - 2)^2$

GRADING NOTE Any equivalent form. a(3)(4) = 12 gives a = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q1/5

A quartic rises at both ends, touches the x-axis at −1 and at 3, and passes through (0, 18). Write the polynomial.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q2/5

675

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$2(x + 1)^2(x - 3)^2$

GRADING NOTE Any equivalent form. a(1)(9) = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q2/5

True or false?

A graph that touches the x-axis at x = a without crossing has a zero of even multiplicity at a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q3/5

676

QUESTION ID

S18-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The factor does not change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q3/5

A cubic falls to the right and crosses the x-axis at 0, 1, and 4. Its leading coefficient is

A. zero

B. not determined by the end behavior

C. positive

D. negative

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q4/5

677

QUESTION ID

S18-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. negative

GRADING NOTE A cubic that falls to the right has a negative leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q4/5

A polynomial graph crosses at −2, touches at 1, falls to the left and rises to the right, and has y-intercept 4. Give a polynomial of least degree.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q5/5

678

QUESTION ID

S18-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2(x + 2)(x - 1)^2$

GRADING NOTE Any equivalent form. Degree 3 with a > 0; a(2)(1) = 4 gives a = 2, which agrees with the end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q5/5

Prove polynomial identities

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Expand the square of a binomial	**HSA-APR.A.1**
BACKWARD 2	Recognize a difference of squares	**HSA-SSE.A.2**
BACKWARD 3	What makes an identity	**6.EE.A.4**
FORWARD 1	Prove or disprove proposed identities	**HSA-APR.C.4**
FORWARD 2	Prove identities with the Binomial Theorem	**HSA-APR.C.5**

SET 19 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Which argument proves the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$?

A. For $x = 2$ and $y = 1$: $25 = 9 + 16$, so the identity is true.

B. Take the square root of both sides: $x^2 + y^2 = (x^2 - y^2) + 2xy$, and the right side simplifies to $x^2 - y^2 + 2xy$, so the identity is true.

C. The left side is $x^4 + 2x^2y^2 + y^4$. The right side is $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The sides are equal.

D. The right side is $(x^2 - y^2)^2 + (2xy)^2 = x^4 - y^4 + 4x^2y^2$, and the left side is $x^4 + y^4 + 4x^2y^2$, so the two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q1/5

681

QUESTION ID
S19-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. The left side is $x^4 + 2x^2y^2 + y^4$. The right side is $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The sides are equal.

GRADING NOTE Both sides expand to $x^4 + 2x^2y^2 + y^4$. The others check one example, square $x^2 - y^2$ as $x^4 - y^4$, or take square roots term by term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q1/5

Which argument proves the identity $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$?

A. The product is $a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$. The terms $-a^2b$ and a^2b cancel, ab^2 and $-ab^2$ cancel, leaving $a^3 + b^3$.

B. For $a = 1$ and $b = 2$ the left side is $1 + 8 = 9$ and the right side is $3 \cdot (1 - 2 + 4) = 9$, so the identity is true for all a and b.

C. Multiply the first terms and the last terms: $a \cdot a^2 = a^3$ and $b \cdot b^2 = b^3$, so the product is $a^3 + b^3$.

D. The product is $a^3 - 2a^2b + b^3$, which equals $a^3 + b^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q2/5

682

QUESTION ID

S19-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. The product is $a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$. The terms $-a^2b$ and a^2b cancel, ab^2 and $-ab^2$ cancel, leaving $a^3 + b^3$.

GRADING NOTE All six products, then the cancellations. The others check one example, skip four of the six products, or multiply incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q2/5

Which argument proves the identity $(a + b)^2 - (a - b)^2 = 4ab$?

A. For $a = 3$ and $b = 1$: $16 - 4 = 12 = 4 \cdot 3$, so the identity is true.

B. $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab$.

C. $(a + b)^2 - (a - b)^2 = [(a + b) - (a - b)]^2 = (2b)^2 = 4ab$.

D. $(a + b)^2 - (a - b)^2 = (a^2 + b^2) - (a^2 - b^2) = a^2 + b^2 - a^2 + b^2 = 4ab$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q3/5

683

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B.

$(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab.$

GRADING NOTE Subtract the whole second square, changing the sign of every term (or factor as $[(a + b) - (a - b)][(a + b) + (a - b)] = (2b)(2a)$). The others check one example, square binomials as $a^2 + b^2$, or use the false rule $x^2 - y^2 = (x - y)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q3/5

Which argument proves the identity $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$?

A. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 + (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

B. For $x = 0$: $4 = 2 \cdot 2$, so the identity is true.

C. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

D. Multiply the first terms and the last terms: $x^2 \cdot x^2 = x^4$ and $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q4/5

684

QUESTION ID

S19-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

GRADING NOTE Group as a difference of squares (or expand all nine products). The others check one example, use $(u + v)(u - v) = u^2 + v^2$, or skip the middle products.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q4/5

Which argument proves the identity
$(n + 1)^3 - n^3 = 3n^2 + 3n + 1$?

A. For n = 1, 2, and 3 the left side is 7, 19, and 37, and so is the right side, so it is true for every n.

B. $(n + 1)^3 = n^3 + 1$, so $(n + 1)^3 - n^3 = 1$, which is $3n^2 + 3n + 1$.

C. $(n + 1)^3 - n^3 = (n + 1 - n)^3 = 1^3 = 1$.

D. $(n + 1)^3 = n^3 + 3n^2 + 3n + 1$, so $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q5/5

685

QUESTION ID
S19-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. $(n + 1)^3 = n^3 + 3n^2 + 3n + 1$, so $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$.

GRADING NOTE Expand $(n + 1)^3$ correctly and subtract. The others check several values (evidence, not a proof) or use the false rules $a^3 - b^3 = (a - b)^3$ and $(n + 1)^3 = n^3 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 1

Expand the square of a binomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q1/5

687

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$a^2 + 2ab + b^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x^2 - y^2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q2/5

688

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^4 - 2x^2y^2 + y^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q2/5

True or false?

$(a - b)^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q3/5

689

QUESTION ID

S19-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(a - b)^2 = a^2 - 2ab + b^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(2xy)^2$ equals

A. $2xy^2$

B. $4xy$

C. $2x^2y^2$

D. $4x^2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q4/5

690

QUESTION ID

S19-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $4x^2y^2$

GRADING NOTE Square each factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q4/5

Expand $(n + 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q5/5

691

QUESTION ID

S19-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$n^3 + 3n^2 + 3n + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 2

Recognize a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $a^2 - b^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q1/5

693

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(a − b)(a + b)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $(x^2 + 2)^2 - (2x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q2/5

694

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 2 - 2x)(x^2 + 2 + 2x)$

GRADING NOTE Any equivalent factored form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q2/5

True or false?

$u^2 - v^2 = (u - v)(u + v)$ for any expressions u and v.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q3/5

695

QUESTION ID

S19-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Structure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q3/5

Which is a difference of squares?

A. $x^3 - 1$

B. $x^2 - 3x$

C. $9x^2 - 16$

D. $9x^2 + 16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q4/5

696

QUESTION ID

S19-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $9x^2 - 16$

GRADING NOTE $(3x)^2 - 4^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q4/5

Factor $25 - x^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q5/5

697

QUESTION ID

S19-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(5 - x^2)(5 + x^2)$

GRADING NOTE Any equivalent factored form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 3

What makes an identity • Grade 6 • Expressions & Equations • 6.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Test $(a + b)^2 = a^2 + b^2$ with $a = 1$ and $b = 2$. Is it an identity?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q1/5

699

QUESTION ID

S19-B3-Q1

TYPE

Short answer

CORRECT ANSWER

No: 9 ≠ 5

GRADING NOTE Form required: the test and the decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain the difference between an equation and an identity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q2/5

700

QUESTION ID

S19-B3-Q2

TYPE

Short answer

CORRECT ANSWER

An identity is true for every value of the variables; an equation may be true only for some values.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q2/5

True or false?

Checking an identity with three numbers proves it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q3/5

701

QUESTION ID

S19-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Examples cannot prove a statement about all values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q3/5

$2(x + 3) = 2x + 6$ is

A. an identity

B. true only for $x = 3$

C. true only for $x = 0$

D. never true

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q4/5

702

QUESTION ID

S19-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. an identity

GRADING NOTE Distributive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q4/5

Give values that show $a^3 + b^3 = (a + b)^3$ is not an identity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q5/5

703

QUESTION ID

S19-B3-Q5

TYPE

Short answer

CORRECT ANSWER

For example

$a = b = 1$: $2 \neq 8$

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 1

Prove or disprove proposed identities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Is $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$ an identity? Which answer is correct?

A. Yes: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, and $3ab(a + b) = 3a^2b + 3ab^2$, so both sides are equal.

B. Yes, because $a = 1$ and $b = 1$ give $8 = 2 + 6$.

C. No: for $a = 1$ and $b = 2$ the left side is $3^3 = 27$ and the right side is $1 + 8 + 3 \cdot 2 = 15$, so the sides differ.

D. No: $(a + b)^3 = a^3 + b^3$, so the extra term makes the two sides different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q1/5

705

QUESTION ID

S19-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Yes: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, and $3ab(a + b) = 3a^2b + 3ab^2$, so both sides are equal.

GRADING NOTE True by expansion. The others use $(a + b)^3 = a^3 + b^3$, check one example, or miscompute ($a = 1$, $b = 2$ gives $27 = 1 + 8 + 18$ on both sides).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q1/5

Is $a^4 - b^4 = (a - b)^4$ an identity? Which answer is correct?

A. Yes: for a = 1 and b = 0 both sides equal 1.

B. No: for a = 2 and b = 1, the left side is 16 − 1 = 15 and the right side is $1^4 = 1$.

C. Yes: a fourth power applies to each term inside the parentheses, so $(a - b)^4 = a^4 - b^4$ for all a and b.

D. No, because $a^4 - b^4 = (a - b)^2(a + b)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q2/5

706

QUESTION ID

S19-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No: for a = 2 and b = 1, the left side is 16 − 1 = 15 and the right side is $1^4 = 1$.

GRADING NOTE One counterexample disproves an identity. The others distribute the exponent, generalize from one example, or use a false factorization ($a^4 - b^4 = (a - b)(a + b)(a^2 + b^2)$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q2/5

True or false?

$(x - 1)(x^2 + x + 1) = x^3 + 1$ is an identity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q3/5

707

QUESTION ID

S19-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The product is $x^3 - 1$ (difference of cubes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q3/5

Which is an identity?

A. $(x + 2)^2 = x^2 + 4$

B. $(x + 1)^3 = x^3 + 1$

C. $x^2 - 4 = (x - 4)^2$

D. $(x + 2)^2 - (x - 2)^2 = 8x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q4/5

708

QUESTION ID

S19-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $(x + 2)^2 - (x - 2)^2 = 8x$

GRADING NOTE 4ab with a = x, b = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find c so that $(x + c)^2 - (x - c)^2 = 12x$ is an identity.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q5/5

709

QUESTION ID

S19-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 4cx = 12x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 2

Prove identities with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Which argument uses the Binomial Theorem to prove that the entries in row n of Pascal's Triangle add to 2^n?

A. Row n has n + 1 entries, and each entry is the sum of the two entries above it, so every row has twice as many entries as the row before it and the sum of row n is 2^n.

B. Row 3 is 1, 3, 3, 1, which adds to $8 = 2^3$, so every row n adds to 2^n.

C. Set x = y = 1 in $(x + y)^n = C(n, 0)x^n + C(n, 1)x^{n-1}y + \ldots + C(n, n)y^n$. The left side is 2^n and the right side is C(n, 0) + C(n, 1) + ... + C(n, n), the row sum.

D. Set x = 1 and y = 0 in $(x + y)^n = C(n, 0)x^n + \ldots + C(n, n)y^n$: the left side is 1 and the right side is C(n, 0) = 1, so each row adds to 2^n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q1/5

711

QUESTION ID
S19-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. Set x = y = 1 in $(x + y)^n = C(n, 0)x^n + C(n, 1)x^{n-1}y + \ldots + C(n, n)y^n$. The left side is 2^n and the right side is C(n, 0) + C(n, 1) + ... + C(n, n), the row sum.

GRADING NOTE Substituting 1 for x and y makes every term equal to its coefficient. The others check one row, use a substitution that gives 1, or argue from the number of entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q1/5

Use the Binomial Theorem with $x = 1$ and $y = -1$ to find $C(5,0) - C(5,1) + C(5,2) - C(5,3) + C(5,4) - C(5,5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q2/5

712

QUESTION ID
S19-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE Any equivalent form. $(1 - 1)^5 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q2/5

True or false?

$(1 + 1)^4 = 1 + 4 + 6 + 4 + 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q3/5

713

QUESTION ID
S19-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q3/5

The sum of the coefficients of $(2x + 1)^3$ is

A. 9

B. 8

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q4/5

714

QUESTION ID
S19-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 27

GRADING NOTE Set x = 1: 3^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $(1 + 0.1)^3 = 1 + 3(0.1) + 3(0.01) + 0.001$ to find 1.1^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q5/5

715

QUESTION ID
S19-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1.331

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q5/5

Use polynomial identities to describe numerical relationships

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Substitute values into an identity	**6.EE.A.2**
BACKWARD 2	Check a Pythagorean triple	**8.G.B.6**
BACKWARD 3	Choose a and b to fit a product	**HSA-APR.C.4**
FORWARD 1	Generate a triple with a required side	**HSA-APR.C.4**
FORWARD 2	Connect the triple identity to complex numbers	**HSN-CN.A.2**

SET 20 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ generates Pythagorean triples. Use $x = 3$ and $y = 2$ to find a triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q1/5

718

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

5, 12, 13

GRADING NOTE Any order. $x^2 - y^2 = 5$, $2xy = 12$, $x^2 + y^2 = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q1/5

Use the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ with $x = 5$ and $y = 2$ to find a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q2/5

719

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

20, 21, 29

GRADING NOTE Any order. $25 - 4 = 21$, $2 \cdot 5 \cdot 2 = 20$, $25 + 4 = 29$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q2/5

Use the identity $(a + b)(a - b) = a^2 - b^2$ to compute 47×53 mentally. Show which a and b you use.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q3/5

720

QUESTION ID

S20-M-Q3

TYPE

Short answer

CORRECT ANSWER

a = 50, b = 3:

2500 − 9 = 2491

GRADING NOTE Form required: the choice of a and b and the product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q3/5

Use the identity $(n + 1)^2 - n^2 = 2n + 1$ to write 41 as a difference of two consecutive squares.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q4/5

721

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

$41 = 21^2 - 20^2$

GRADING NOTE Form required: the two squares. $2n + 1 = 41$ gives $n = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q4/5

Use the identity $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$ to compute $13^3 - 12^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q5/5

722

QUESTION ID
S20-M-Q5

TYPE
Short answer

CORRECT ANSWER
469

GRADING NOTE Any equivalent form. 3·144 + 36 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 1

Substitute values into an identity • Grade 6 • Expressions & Equations • 6.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $x^2 - y^2$ for $x = 3$ and $y = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q1/5

724

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2xy for x = 5 and y = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q2/5

725

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q2/5

True or false?

For $n = 12$, $3n^2 + 3n + 1 = 433$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q3/5

726

QUESTION ID

S20-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $432 + 36 + 1 = 469$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $a = 50$ and $b = 3$, $a^2 - b^2$ is

A. 2491

B. 2500

C. 2494

D. 2509

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q4/5

727

QUESTION ID

S20-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2491

GRADING NOTE 2500 − 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2n + 1 for n = 20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q5/5

728

QUESTION ID

S20-B1-Q5

TYPE

Short answer

CORRECT ANSWER

41

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 2

Check a Pythagorean triple • Grade 8 • Geometry • 8.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Do 5, 12, 13 satisfy $a^2 + b^2 = c^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q1/5

730

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: 25 + 144 = 169

GRADING NOTE Form required: the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Do 20, 21, 29 satisfy $a^2 + b^2 = c^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q2/5

731

QUESTION ID

S20-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes:

400 + 441 = 841

GRADING NOTE Form required: the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q2/5

True or false?

6, 8, 11 is a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q3/5

732

QUESTION ID

S20-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $36 + 64 = 100 \neq 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q3/5

Which is a Pythagorean triple?

A. 8, 15, 17

B. 6, 7, 9

C. 2, 3, 4

D. 4, 5, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q4/5

733

QUESTION ID

S20-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8, 15, 17

GRADING NOTE 64 + 225 = 289.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q4/5

Explain why a triangle with sides 5, 12, 13 is a right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q5/5

734

QUESTION ID

S20-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$5^2 + 12^2 = 13^2$, so by the converse of the Pythagorean Theorem it is a right triangle.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 3

Choose a and b to fit a product • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

ANSWER KEY Reduced copy of the question page, with its answer

Write 47 and 53 as 50 − b and 50 + b. What is b?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q1/5

736

QUESTION ID

S20-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q1/5

Write 98 × 102 as (a − b)(a + b).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q2/5

737

QUESTION ID

S20-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(100 − 2)(100 + 2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$29 \times 31 = 30^2 + 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q3/5

738

QUESTION ID

S20-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(30 - 1)(30 + 1) = 30^2 - 1 = 899.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q3/5

64 × 56 equals

A. $(60 - 4)^2$

B. $64^2 - 56^2$

C. $60^2 - 4^2$

D. $60^2 + 4^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q4/5

739

QUESTION ID

S20-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $60^2 - 4^2$

GRADING NOTE a = 60, b = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 98 × 102.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q5/5

740

QUESTION ID

S20-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9996

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 1

Generate a triple with a required side • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Use the triple identity to find x and y (whole numbers, x > y) that give a triple with hypotenuse 25 and legs 7 and 24.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q1/5

742

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 4, y = 3

GRADING NOTE Any equivalent form. $x^2 + y^2 = 25$, $x^2 - y^2 = 7$, $2xy = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find x and y (whole numbers, x > y) for which the identity gives a triple with a leg 2xy = 40 and hypotenuse 41.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q2/5

743

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 5, y = 4

GRADING NOTE Any equivalent form. $x^2 - y^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q2/5

True or false?

Every choice of whole numbers $x > y > 0$ in the identity gives a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q3/5

744

QUESTION ID

S20-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All three expressions are positive whole numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q3/5

Which choice gives the triple 8, 15, 17?

A. x = 4, y = 1

B. x = 5, y = 3

C. x = 4, y = 2

D. x = 3, y = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q4/5

745

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x = 4, y = 1

GRADING NOTE 16 − 1 = 15, 8, 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q4/5

Explain why $x = 2y$ gives a triple with legs in the ratio $3 : 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q5/5

746

QUESTION ID

S20-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^2 - y^2 = 3y^2$ and $2xy = 4y^2$, so the legs are $3y^2$ and $4y^2$ (ratio 3 : 4).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 2

Connect the triple identity to complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(3 + 2i)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q1/5

748

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5 + 12i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(x + yi)^2$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q2/5

749

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x^2 - y^2) + 2xyi$

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q2/5

True or false?

The real and imaginary parts of $(x + yi)^2$ are the legs of the Pythagorean triple from the identity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q3/5

750

QUESTION ID
S20-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $x^2 - y^2$ and $2xy$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q3/5

$(5 + 2i)^2$ equals

A. 25 + 4i

B. 21 + 20i

C. 21 + 10i

D. 29 + 20i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q4/5

751

QUESTION ID

S20-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 21 + 20i

GRADING NOTE 25 − 4 + 20i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q4/5

Use $(4 + i)^2$ to find a Pythagorean triple.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q5/5

752

QUESTION ID

S20-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15, 8, 17

GRADING NOTE Any order. $(4 + i)^2 = 15 + 8i$; $4^2 + 1^2 = 17$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q5/5

Expand with the Binomial Theorem

MAIN	Algebra II main question	**HSA-APR.C.5**
BACKWARD 1	Multiply binomials repeatedly	**HSA-APR.A.1**
BACKWARD 2	Build Pascal's Triangle by a rule	**4.OA.C.5**
BACKWARD 3	Powers of products and of negatives	**8.EE.A.1**
FORWARD 1	Find a specific term or coefficient with a condition	**HSA-APR.C.5**
FORWARD 2	Use binomial coefficients to compute probabilities	**HSS-CP.B.9**

SET 21 • HSA-APR.C.5

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Use the Binomial Theorem to expand $(x + 2)^5$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q1/5

755

QUESTION ID
S21-M-Q1

TYPE
Short answer

CORRECT ANSWER
$x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$

GRADING NOTE Form required: standard form. Coefficients $C(5, k)\cdot 2^k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q1**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q1/5

Use the Binomial Theorem to expand $(2a - b)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q2/5

756

QUESTION ID
S21-M-Q2

TYPE
Short answer

CORRECT ANSWER
$16a^4 - 32a^3b + 24a^2b^2 - 8ab^3 + b^4$

GRADING NOTE Any order of terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q2**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q2/5

Use the Binomial Theorem to find the coefficient of x^3 in $(x + 3)^7$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q3/5

757

QUESTION ID
S21-M-Q3

TYPE
Short answer

CORRECT ANSWER
2835

GRADING NOTE Any equivalent form. $C(7, 3)\cdot 3^4 = 35\cdot 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q3/5

Use the Binomial Theorem to find the term containing x^3y^3 in $(x - 2y)^6$.

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q4/5

758

QUESTION ID
S21-M-Q4

TYPE
Short answer

CORRECT ANSWER
$-160x^3y^3$

GRADING NOTE Any equivalent form. $C(6, 3)(-2)^3 = 20\cdot(-8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q4**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q4/5

Use the first three terms of the binomial expansion of $(1 + 0.02)^5$ to estimate 1.02^5 to four decimal places.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q5/5

759

QUESTION ID
S21-M-Q5

TYPE
Short answer

CORRECT ANSWER
1.1040

GRADING NOTE Any value 1.1040 or 1.1041 is accepted (1 + 0.1 + 0.004 = 1.104; the exact value is 1.10408...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 1

Multiply binomials repeatedly • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q1/5

761

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 4x + 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q2/5

762

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + 6x^2 + 12x + 8$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q2/5

True or false?

$(x + y)^3 = x^3 + y^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q3/5

763

QUESTION ID

S21-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$; for $x = y = 1$, $8 \neq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The number of terms in the expansion of $(x + y)^5$ is

A. 6

B. 5

C. 10

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q4/5

764

QUESTION ID

S21-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE n + 1 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(a - b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q5/5

765

QUESTION ID

S21-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$a^2 - 2ab + b^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 2

Build Pascal's Triangle by a rule • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Row 4 of Pascal's Triangle is 1, 4, 6, 4, 1. Write row 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q1/5

767

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1, 5, 10, 10, 5, 1

GRADING NOTE Form required: all six entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q1/5

Write row 7 of Pascal's Triangle (row 6 is 1, 6, 15, 20, 15, 6, 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q2/5

768

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1, 7, 21, 35, 35, 21, 7, 1

GRADING NOTE Form required: all entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q2/5

True or false?

Each inner entry of Pascal's Triangle is the sum of the two entries above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q3/5

769

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The rule of the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q3/5

The fourth entry of row 7 (counting from the first 1) is

A. 20

B. 7

C. 35

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q4/5

770

QUESTION ID

S21-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 35

GRADING NOTE 1, 7, 21, 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the sum of the entries in row 5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q5/5

771

QUESTION ID

S21-B2-Q5

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 3

Powers of products and of negatives • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(2a)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q1/5

773

QUESTION ID

S21-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$16a^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(-2y)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q2/5

774

QUESTION ID

S21-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$-8y^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q2/5

True or false?

$(-b)^4 = -b^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q3/5

775

QUESTION ID

S21-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An even power of $-b$ is positive: $(-b)^4 = b^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

3^4 equals

A. 27

B. 12

C. 64

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q4/5

776

QUESTION ID

S21-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 81

GRADING NOTE 3·3·3·3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(0.02)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q5/5

777

QUESTION ID

S21-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.0004

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 1

Find a specific term or coefficient with a condition • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Find the constant term of $(x + 1/x)^6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q1/5

779

QUESTION ID
S21-F1-Q1

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE Any equivalent form. Middle term C(6, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q1/5

The coefficient of x^2 in $(1 + kx)^5$ is 90. Find k ($k > 0$).

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q2/5

780

QUESTION ID
S21-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. $C(5, 2)k^2 = 10k^2 = 90$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q2/5

True or false?

The middle term of $(x + y)^8$ is $56x^4y^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q3/5

781

QUESTION ID

S21-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $C(8, 4) = 70$, so the middle term is $70x^4y^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q3/5

The term with x^4 in $(x^2 - 1)^5$ is

A. $-5x^4$

B. $-10x^4$

C. $10x^4$

D. $5x^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q4/5

782

QUESTION ID
S21-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $-10x^4$

GRADING NOTE $C(5, 2)(x^2)^2(-1)^3 = -10x^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q4/5

Find the coefficient of x in $(2x - 1)^6$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q5/5

783

QUESTION ID
S21-F1-Q5

TYPE
Short answer

CORRECT ANSWER
−12

GRADING NOTE Any equivalent form. $C(6, 1)\cdot 2\cdot(-1)^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 2

Use binomial coefficients to compute probabilities • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is tossed 5 times. Use C(5, k) to find P(exactly 3 heads).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q1/5

785

QUESTION ID
S21-F2-Q1

TYPE
Short answer

CORRECT ANSWER
10/32 = 5/16

GRADING NOTE Any equivalent form (0.3125).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner lands on red with probability 0.3. Find P(exactly 2 reds in 4 spins).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q2/5

786

QUESTION ID
S21-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.2646

GRADING NOTE Any equivalent form. $C(4, 2)(0.3)^2(0.7)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q2/5

True or false?

The probabilities of 0, 1, ..., n successes add to 1 because $(p + (1 - p))^n = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q3/5

787

QUESTION ID
S21-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Binomial Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For 6 fair coin tosses, P(exactly 3 heads) is

A. 6/64

B. 3/6

C. 15/64

D. 20/64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q4/5

788

QUESTION ID

S21-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20/64

GRADING NOTE C(6, 3) = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q4/5

Explain why C(n, k) counts the outcomes with exactly k successes in n trials.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q5/5

789

QUESTION ID
S21-F2-Q5

TYPE
Short answer

CORRECT ANSWER

Each outcome is a choice of which k of the n trials are successes; there are C(n, k) such choices.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q5**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q5/5

Rewrite rational expressions by polynomial division

MAIN	Algebra II main question	**HSA-APR.D.6**
BACKWARD 1	Write a division with a remainder as a mixed number	**4.NF.B.3**
BACKWARD 2	Multiply to check a division	**HSA-APR.A.1**
BACKWARD 3	Use the Remainder Theorem to check the remainder	**HSA-APR.B.2**
FORWARD 1	Divide with unknown coefficients and higher-degree divisors	**HSA-APR.D.6**
FORWARD 2	Use division to find slant asymptotes	**HSF-IF.C.7.d**

SET 22 • HSA-APR.D.6

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Which shows $(x^3 + 2x^2 - 5x + 3)/(x - 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $x^2 + x - 6 + 9/(x - 1)$

B. $x^2 + 3x - 2 + 1/(x + 1)$

C. $x^2 + 3x - 2 - 1/(x - 1)$

D. $x^2 + 3x - 2 + 1/(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q1/5

792

QUESTION ID
S22-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. $x^2 + 3x - 2 + 1/(x - 1)$

GRADING NOTE Synthetic division by 1: 1, 3, −2, remainder 1. The others divide by −1 instead of 1 ($x^2 + x - 6$, remainder 9), change the sign of the remainder, or change the divisor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q1/5

Which shows $(2x^3 - 3x^2 + 4x - 7)/(x^2 + 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $2x - 3 + (2x + 4)/(x^2 + 1)$

B. $2x - 3 + (2x - 10)/(x^2 + 1)$

C. $2x - 3 + (2x - 4)/(x^2 + 1)$

D. $2x - 3 + (6x - 10)/(x^2 + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q2/5

793

QUESTION ID

S22-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C.

$2x - 3 + (2x - 4)/(x^2 + 1)$

GRADING NOTE $(x^2 + 1)(2x - 3) = 2x^3 - 3x^2 + 2x - 3$; subtracting from the dividend leaves $2x - 4$ (degree less than 2). The others add instead of subtracting, or make a sign error in one term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q2/5

Which shows $(x^4 - 16)/(x - 2)$ written in the form $q(x) + r(x)/b(x)$?

A. $x^3 + 2x^2 + 4x + 8$ (remainder 0)

B. $x^3 + 2x^2 + 4x + 8 - 32/(x - 2)$

C. $x^3 - 2x^2 + 4x - 8$ (remainder 0)

D. $x^3 + 8$ (remainder 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q3/5

794

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $x^3 + 2x^2 + 4x + 8$ (remainder 0)

GRADING NOTE Synthetic division by 2 with coefficients 1, 0, 0, 0, −16: 1, 2, 4, 8, remainder 0. The others divide by x + 2, leave out the zero placeholders, or miscompute the remainder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q3/5

Which shows $(6x^2 + x - 4)/(2x + 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $6x - 2 - 3/(2x + 1)$

B. $3x - 1 - 4/(2x + 1)$

C. $3x - 1 + 3/(2x + 1)$

D. $3x - 1 - 3/(2x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q4/5

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QUESTION ID

S22-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D.

3x − 1 − 3/(2x + 1)

GRADING NOTE $(2x + 1)(3x - 1) = 6x^2 + x - 1$, so the remainder is $-4 - (-1) = -3$. The others change the sign of the remainder, use synthetic division by $-1/2$ without dividing the quotient by 2, or forget to subtract the -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q4/5

Which shows $(x^3 - 4x + 5)/(x + 3)$ written in the form $q(x) + r(x)/b(x)$?

A. $x - 7 + 26/(x + 3)$

B. $x^2 - 3x + 5 - 10/(x + 3)$

C. $x^2 - 3x + 5 + 10/(x + 3)$

D. $x^2 + 3x + 5 + 20/(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q5/5

796

QUESTION ID

S22-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B.

$x^2 - 3x + 5 - 10/(x + 3)$

GRADING NOTE Synthetic division by -3 with coefficients 1, 0, -4, 5: 1, -3, 5, remainder -10. The others leave out the 0 for x^2, divide by 3 instead of -3, or change the sign of the remainder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 1

Write a division with a remainder as a mixed number • Grade 4 • Number & Operations–Fractions • 4.NF.B.3

Write 23/4 as a mixed number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q1/5

798

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5 3/4

GRADING NOTE Any equivalent form (5 + 3/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 47/5 in the form q + r/5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q2/5

799

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9 + 2/5

GRADING NOTE Form required: q + r/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

17/3 = 5 + 1/3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q3/5

800

QUESTION ID

S22-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 17 = 3 · 5 + 2, so 17/3 = 5 + 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q3/5

800